

Can the Public Carry Trade Restore Fiscal Sustainability?

Evidence from Korea's Consolidated Balance Sheet*

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Abstract

Can governments use a carry-trade strategy to offset the fiscal costs of population aging? We examine this question through the consolidated balance sheet of Korea's public sector, combining the assets and liabilities of the government, public pension and investment funds, and the central bank. The consolidated balance sheet reveals increasing reliance on a carry trade in which risky, increasingly foreign assets are financed with safe, low-yielding domestic liabilities. This strategy has generated a favorable return spread and strengthened public net worth. However, present-value accounting shows that projected government and pension deficits substantially exceed the public sector's accumulated financial wealth. A VAR-based simulation similarly indicates that persistent structural deficits eventually overwhelm the returns from public assets and drive consolidated net worth deeply negative. Financial repression can delay this deterioration but cannot restore fiscal sustainability; moreover, it shifts the cost disproportionately to less financially sophisticated depositors. Neither the carry trade nor financial repression can substitute for fiscal and pension reform.

JEL Classification: E58, E62, E63, G11, G12, H63.

Keywords: Consolidated Balance Sheet, Carry Trade, Financial Repression, Fiscal Sustainability.

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1 Introduction

Many developed economies are undergoing rapid population aging and, in some cases, population decline. These demographic shifts are likely to slow economic growth while increasing pension, health-care, and other age-related expenditures. Governments may consequently face large and persistent fiscal deficits. At the same time, central banks and other public financial institutions have substantially expanded their balance sheets and, in many cases, increased their exposure to market risk over the past several decades. Evaluating fiscal sustainability therefore requires looking beyond government debt alone to the assets, liabilities, and risks of the consolidated public sector.

Korea brings these two developments into especially sharp focus. It is experiencing one of the fastest demographic transitions among developed economies, while its public sector holds large portfolios with substantial exposure to risky domestic and foreign assets through the National Pension Service (NPS), the Korea Investment Corporation (KIC), and the Bank of Korea (BoK). Although Korea's gross government debt remains moderate, this combination raises a broader question: can the returns on public assets offset the persistent deficits associated with population aging, and who ultimately bears the risks of this strategy?

We address these questions by constructing a consolidated balance sheet for Korea's public sector. We combine the central and local governments, the National Pension Service, the Korea Investment Corporation, and the Bank of Korea, and measure both the value and the returns of their assets and liabilities. Consolidation is economically appropriate because the profits and losses of these publicly owned institutions ultimately accrue to the public and because the monetary and fiscal authorities face a common intertemporal budget constraint ([Sargent and Wallace, 1981](#); [Hall and Sargent, 2022](#)). The relevant measure of fiscal capacity is therefore the net worth of the consolidated public sector—the value of its financial assets net of its liabilities—rather than the gross debt of the central government alone.

Viewed through the consolidated balance sheet, Korea's fiscal position initially appears stronger than its gross-debt figures suggest. The public sector's net asset position reached 45.9% of GDP in 2024, an improvement of 17.6 percentage points since 2015, while the NPS alone held assets equal to approximately 46% of GDP. The composition of the balance sheet, however, reveals substantial risk-taking: public assets have increasingly shifted toward risky securities, especially foreign equities, while liabilities consist largely of safe, low-yielding domestic instruments such as government bonds and monetary-policy liabilities. Reflecting this composition, net investment income—asset returns less funding costs, scaled by GDP—increased from an average of approximately 1.5% of GDP over 2008–2015 to 4.3% over 2015–2024, as the return spread between assets and liabilities rose from 0.2 to 3.5 percentage points. We refer to this combination of risky, higher-return assets and safe, lower-cost liabilities as a public-sector carry trade. This position was not the product of a coordinated investment strategy, but emerged from the separate mandates of the NPS, the KIC,

and other public institutions.

The public-sector carry trade, however, creates a fundamental risk mismatch. The government's pension promises and debt obligations are relatively safe, while the assets backing them are exposed to aggregate risk and may lose value precisely when the domestic economy is weak. The higher expected return on these assets is therefore compensation for risk rather than a new fiscal resource. Under the consolidated government budget constraint, this risk must ultimately be borne either by bondholders, through higher yields or losses, or by taxpayers, through higher taxes and reduced public services in adverse economic states (Jiang et al., 2024). The relevant question is therefore whether the expected return on Korea's public assets is sufficient to offset its demographic deficits once the associated risk is recognized.

To determine whether this return-seeking strategy can deliver fiscal sustainability, we use two complementary exercises. First, a present-value accounting exercise compares the discounted value of projected central-government and NPS deficits with the public sector's current net asset position. Second, a VAR-based Monte Carlo simulation projects the consolidated balance sheet under stochastic asset returns, funding costs, economic growth, and fiscal deficits. This exercise characterizes the timing and uncertainty of the deterioration in public net worth.

The present-value accounting reveals a substantial imbalance. NABO's projections imply that the present value of central-government primary deficits alone through 2072 is approximately 100% of current GDP—more than twice the public sector's current net asset position of 45.9% of GDP. Including projected NPS deficits further raises the present value of total deficits to more than 164% of current GDP. After accounting for current public assets, the remaining financing gap is at least 120% of GDP and must ultimately be closed through fiscal adjustment or higher-than-projected asset returns.

The stochastic simulation reaches the same conclusion. Although the public-sector carry trade continues to earn a median return spread of approximately 1.9 percentage points, persistent structural deficits eventually overwhelm these gains. The consolidated net asset position turns negative around 2061, the NPS fund is exhausted around 2070, and public net worth falls to approximately −76% of GDP by 2072. A Bayesian-VAR robustness exercise moderates the terminal decline to approximately −59% of GDP but does not overturn the result. Public investment returns therefore delay, but do not prevent, the deterioration of Korea's fiscal position.

We next ask whether financial repression can provide fiscal relief when the carry trade cannot. By financial repression, we mean financing the government at below-market rates through captive public or regulated institutions. We consider both a balance-sheet policy, in which additional government financing is provided through low-cost central-bank reserves, and a quasi-fiscal policy that requires captive institutions to hold public debt at below-market rates. We vary the timing and intensity of these policies while holding economic growth and asset returns fixed, thereby giving financial repression a favorable evaluation.

Financial repression delays but does not eliminate Korea's fiscal shortfall. The balance-sheet

policy has little effect, improving the projected 2072 net asset position only from -76% to -73% of GDP. An early quasi-fiscal intervention with a -200 -basis-point wedge improves the terminal position to -31% of GDP but does not restore solvency. Only an extreme -400 -basis-point wedge brings terminal net worth close to balance, at approximately -5% of GDP. The NPS fund is nevertheless exhausted around 2070 under every scenario. Because the simulations hold growth and asset returns fixed, these estimates are upper bounds on the fiscal benefits of repression.

Financial repression also creates no new aggregate resources. The below-market return forgone by captive creditors is transferred to the government and ultimately borne disproportionately by depositors, especially less financially sophisticated households. The risk-premium identity for the consolidated balance sheet implies that this burden becomes increasingly large as public liabilities rise relative to net worth. Financial repression therefore redistributes the cost of fiscal adjustment rather than removing it. Like public-sector risk-taking, it may buy time, but it cannot substitute for fiscal and pension reform.

Our paper uses Korea to evaluate whether public-sector risk-taking can expand fiscal capacity enough to absorb the costs of population aging. It makes four specific contributions. First, we construct, to our knowledge, the first consolidated public-sector balance sheet and return series for Korea, measuring fiscal capacity through public net worth rather than gross government debt alone. Second, we document that Korea's public sector conducts a carry trade by holding risky assets while issuing safe, low-yielding liabilities, and we identify how the associated aggregate risk is allocated. Third, we quantify whether this strategy can absorb Korea's projected demographic deficits using both present-value accounting and stochastic simulations. Fourth, we evaluate financial repression as an alternative policy and show that it can delay fiscal adjustment only by transferring the burden to captive creditors, especially less financially sophisticated depositors.

Related literature. Our paper first contributes to research that evaluates government finances using a consolidated public-sector balance sheet. [Buiter \(1985\)](#) develops a comprehensive measure of public net worth, while [International Monetary Fund \(2018\)](#) and [Chai et al. \(2024\)](#) show why public assets should be incorporated into fiscal measurement and policy. A related asset-pricing literature measures the returns and risks on both sides of the public balance sheet. [Chien et al. \(2025a\)](#) apply this approach to Japan, and related studies examine the Eurozone and Taiwan ([Chien et al., 2025b](#); [Chang et al., 2025](#)). We apply this framework to Korea and use it to test whether returns on public wealth can offset the fiscal consequences of population aging.

Second, our paper contributes to the literature on fiscal sustainability and fiscal space. Present-value and generational-accounting approaches compare current resources with future government obligations ([Auerbach et al., 1994](#); [Bohn, 1998](#)), while the fiscal-space literature measures the distance between current debt and an endogenous debt limit ([Heller, 2005](#); [Ostry et al., 2010](#); [Ghosh et al., 2013](#)). More recent research emphasizes that government surpluses are risky cash flows and therefore require risk-adjusted valuation ([Jiang et al., 2024, 2022, 2020](#)). We compare

Korea’s projected deficits with an asset-inclusive measure of fiscal capacity: the net worth of the consolidated public sector.

Third, our analysis relates to research on the safety and liquidity services provided by government liabilities. Investors’ demand for these services allows governments to borrow at rates below the return on risky capital ([Krishnamurthy and Vissing-Jorgensen, 2012](#); [Reis, 2022](#)). This funding advantage is central to the debate over fiscal sustainability when the safe interest rate is below economic growth ([Blanchard, 2019](#); [Barro, 2023](#); [Mian et al., 2025](#)). We evaluate the government’s funding advantage jointly with the returns and aggregate risk on its assets, thereby interpreting Korea’s consolidated position as a public-sector carry trade.

Fourth, our policy analysis contributes to research on financial repression. [Reinhart and Sbrancia \(2015\)](#) document its role in reducing postwar public debt, while [Jeanne \(2025\)](#) studies its optimal timing. Related work measures the funding benefits and limits of below-market government financing ([Chien et al., 2025a](#); [Reis, 2022, 2025](#)). We quantify whether repression can restore Korea’s fiscal sustainability and use the consolidated risk-premium identity to identify who ultimately bears its cost.

Finally, our paper relates to research on Korea’s fiscal and pension outlook. Government and institutional studies project a rapid increase in public debt under population aging ([NABO, 2025](#); [Kim, 2022](#); [Kang and Han, 2023](#)), while related work emphasizes the political difficulty of pension reform ([Lee et al., 2023](#); [Lee and Shin, 2024](#); [Alesina and Drazen, 1991](#)). These studies focus primarily on projected deficits and government liabilities. We incorporate the public sector’s assets, investment returns, and aggregate risk through a consolidated balance-sheet analysis.

The rest of the paper is organized as follows. Section 2 briefly describes the dynamics of Korea’s fiscal deficit over the past decades and highlights the dim fiscal outlook caused by a rapidly aging society. The government’s risk-taking investment strategy is documented in Section 3, which presents the consolidated balance sheet and its estimated returns since 2008. Section 4 offers a simple theoretical and quantitative analysis to assess the sustainability of Korea’s public finances under the risky investment strategy. Section 5 asks whether financial repression could restore sustainability where the carry trade cannot. Section 6 concludes the study.

2 Korea’s Fiscal Dynamics

This section sets out the fiscal challenge that motivates the rest of the paper. Its message is twofold: Korea’s public finances have weakened over time—with a pronounced deterioration since the COVID-19 pandemic—and, looking ahead, exceptionally rapid population aging is likely to deepen the strain further. The first subsection reviews the historical trends in the primary balances of the central government, local governments, and the social security fund. The second documents Korea’s demographic outlook—its rapidly rising elderly share, shrinking population, and slowing trend growth. The third presents the National Assembly Budget Office’s (NABO)

long-term fiscal projections under this demographic pressure.

2.1 Historical Trends

Figure 1 plots the primary fiscal surplus or deficit as a percentage of GDP for the central government, local governments, and the social security fund.¹ The central government has run a primary deficit in most years—20 out of the 25 years in the sample (2001–2025). In contrast, local government finances have generally maintained surpluses. The social security fund has consistently run a primary surplus throughout the sample period, though this surplus has narrowed to near zero by 2025.

Following the 1997 Asian financial crisis, central government fiscal balances deteriorated due to expanded fiscal spending for economic recovery and financial sector restructuring. The year 2003 marked a particularly severe deterioration in central government finances, recording one of the largest deficits in the sample period. This fiscal shock resulted from three simultaneous pressures: the credit card crisis requiring extensive public fund interventions for major issuers like LG Card and Samsung Card, Typhoon Maemi damage necessitating emergency disaster recovery spending, and the beginning of systematic public fund repayment obligations under the newly established legal framework. These factors combined to create an unprecedented fiscal strain on the central budget.

From 2004 through 2006, central government deficits persisted primarily due to ongoing public fund repayment burdens. The Public Fund Repayment Fund Act mandated substantial annual debt service through government bond issuance, creating sustained fiscal drag. Furthermore, demographic pressures accelerated social welfare spending growth, with outlays rising steadily and increasing their share of total central expenditures during this period.

The central government experienced a dramatic fiscal turnaround in 2007, recording a substantial surplus for the first time in several years. This improvement stemmed from three key factors: the conclusion of the multiyear public fund bond conversion cycle (2003–2006), which reduced annual debt service burdens; corporate and personal tax revenues that far exceeded projections due to robust corporate profits and active real estate transactions; and strong export performance supported by favorable exchange rate conditions.

However, the 2008–2009 global financial crisis quickly reversed these gains. The government enacted substantial supplementary budgets to fund bank rescue measures, export support, and domestic stimulus programs. At the same time, the private sector contraction sharply reduced

¹These figures are taken from the Bank of Korea's national accounts (an SNA, accrual basis), which differ conceptually from the Ministry of Economy and Finance's budgetary statistics (a cash basis built around the managed fiscal balance) in both timing and the treatment of social security funds and property income. The two sources can therefore disagree on which year records the largest central-government deficit: on the national-accounts basis used here the deepest post-2000 deficit is 2003 (the credit-card crisis and the onset of public-fund repayments), whereas under the MOEF managed-balance statistics it is 2020 (the COVID-19 response). The primary balance is measured as the total surplus net of net property income, i.e. total revenue less total expenditure less net interest and other property income.

tax receipts, creating severe revenue-expenditure imbalances that pushed central government balances deep into deficit territory.

From 2008 to 2013, central government deficits persisted despite economic recovery efforts. A policy of pursuing stimulus without tax increases—including corporate and personal tax reductions—constrained revenue growth even as spending pressures mounted.

From 2013 to 2017, temporary fiscal improvement was achieved through tax base expansion rather than rate increases. By converting numerous income deductions into tax credits while maintaining headline tax rates, the effective tax base was broadened, generating revenue surpluses in 2016–2017. This period also benefited from a rebound in corporate profits amid favorable export conditions and from structural adjustments to discretionary spending that restrained expenditure growth. This combination of a broader tax base and controlled spending produced Korea’s first sustained central government surpluses since the 2008 crisis.

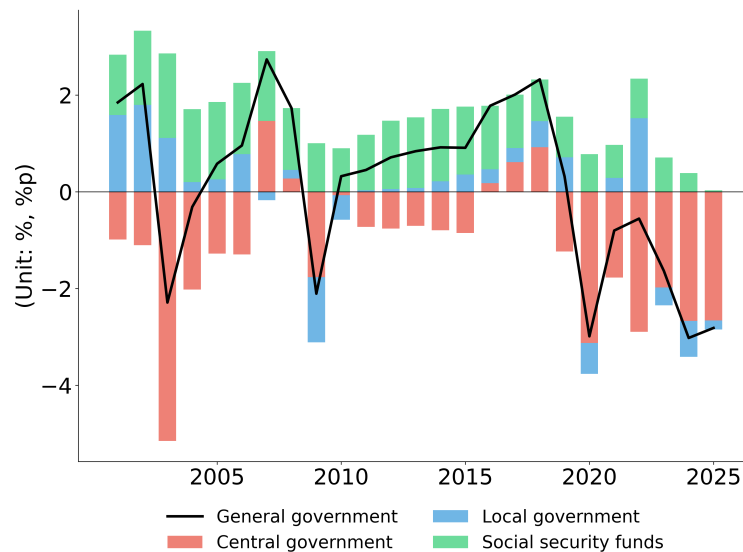
From 2020 onward, the COVID-19 pandemic introduced unprecedented fiscal pressures. The government enacted multiple successive supplementary budgets for health spending, emergency income transfers, and business support programs. These emergency measures, combined with pandemic-induced revenue shortfalls, pushed the central-government primary balance to about -3.1% of GDP in 2020—among the deepest deficits of the sample, though still short of the 2003 trough (-5.2%) on the national-accounts basis used here. The fiscal position deteriorated again at the end of the sample: a sharp revenue shortfall widened the central-government primary deficit to -2.7% of GDP in 2024, leaving the general government at -3.0% .

Local governments in South Korea have generally remained in relatively good fiscal shape. In 2000, the central government increased the share of national tax revenues allocated to local governments from 13.27% to 15%, significantly boosting local revenues through a larger portion of national tax collections. As a result, local governments have typically been able to balance their budgets and even run modest surpluses during normal periods. The two major exceptions were during the global financial crisis (2009–2010) and the COVID-19 pandemic (2020), when fiscal pressures led to temporary deficits.

2.2 Fast Aging Population

As the previous subsection showed, Korea’s fiscal position has weakened over the past two decades, and the deterioration since the COVID-19 pandemic has been especially pronounced. Looking ahead, the single most important force that threatens to entrench and deepen these deficits is demographic. In many developed countries, such as Japan and Italy, the population is currently aging and shrinking. Korea will be no exception in this regard, but the speed of aging faced ahead by Korean society is much faster compared to other countries. Figure 2a plots the actual and projected share of the elderly population (65 or above) for Korea and Japan in the first half of the 21st century. Japan has been at the forefront of the development of an aging society. In 2001, Japan’s share of the aged population was 18.4%, which matches Korea’s recent level in 2023. However,

Figure 1: Government Primary Surplus as a Percentage of GDP



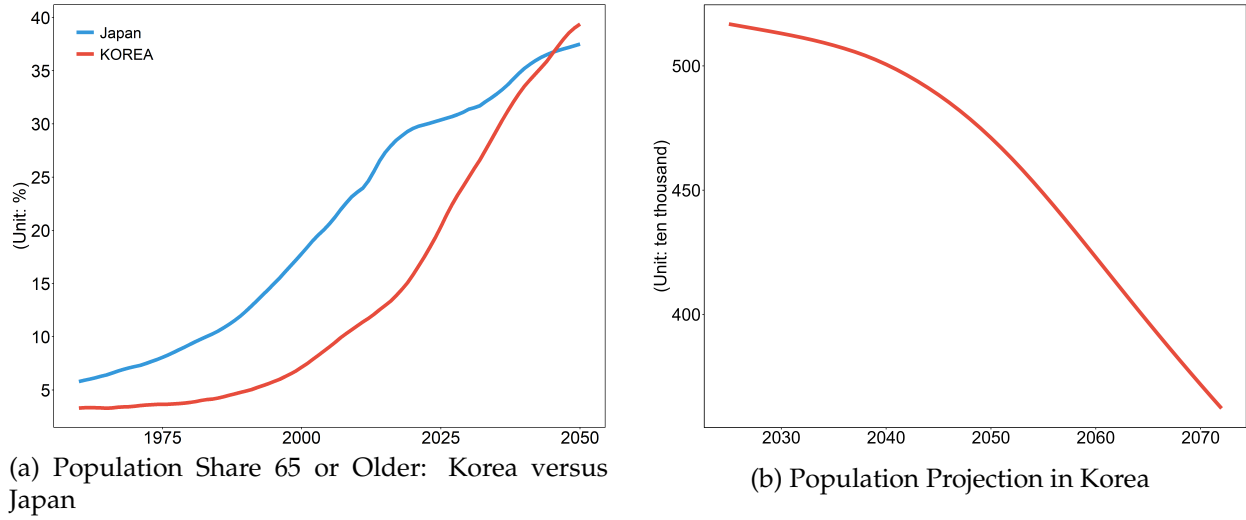
Note: The primary balance is defined as the total surplus net of property income.

Source: ECOS (Bank of Korea, General Government and Public Sector Account).

by 2046, the share of the aged population in both countries will be the same, reaching 37%. In other words, it took Japan 45 years to double its share of the aged population from 18.4% to 37%, while Korea will only take half the time (23 years) to achieve the same increase. Additionally, it is clear from the projections that the long-term level of the aged population will be even higher in Korea than in Japan. In short, Korea has begun the process of becoming an aging society. More importantly, the speed of aging is about twice as fast as that of its nearby neighbor, Japan, with an even higher share of the aged population after three decades. South Korea is now experiencing one of the fastest demographic transitions in the world, with its total population shrinking and its elderly share soaring. As shown in Figure 2b, population projections from KOSIS indicate that Korea's population is projected to fall from about 51.5 million in 2025 to below 36 million by 2072.

This exceptionally rapid aging does not remain confined to demography; it reshapes the macroeconomic environment in which fiscal policy operates. A shrinking, older population erodes the labor force and drags down potential growth, narrowing the tax base even as age-related spending rises. According to NABO's long-term macroeconomic outlook, South Korea's real GDP growth is projected to slow sharply as the population contracts and ages (Table 1). Growth falls from 2.2 percent in 2025 to 1.2 percent by 2040, 0.8 percent in 2050, and just 0.3 percent by 2072. Consumer inflation eases to the Bank of Korea's 2 percent target by 2026 and remains there, despite weaker demand. Nominal wage growth rises from 2.5 percent in 2023 to about 3.9 percent by 2028 and then hovers around 3.7–4.0 percent (3.8 percent on average). The 3-year government bond yield falls from 3.1 percent in 2024 to 2.5 percent in 2025 and then stabilizes around 2.1–2.2 percent,

Figure 2: Korea's Population projection



Note: The figure is based on the medium-variant projection. The elderly population shares for both Korea and Japan are taken from the World Bank to keep the cross-country comparison on a single, consistent source and definition. Statistics Korea's projection (KOSIS) gives a somewhat higher path for Korea — a 65-and-over share of 39.8% in 2046 versus about 37% here — but this does not affect the comparison, which is that Korea is aging at roughly twice the speed Japan did.

Source: World Bank [Population Estimates and Projections](#); KOSIS (Statistics Korea) for the Korean figure cited in the note.

reflecting low potential growth and anchored inflation.²

Table 1: NABO Long-Term Fiscal Outlook: Key Macroeconomic Variables (Medium Population Scenario)

Variable	2025	2030	2040	2050	2060	2072	Average
Real GDP Growth Rate	2.2%	1.9%	1.2%	0.8%	0.5%	0.3%	1.0%
Nominal GDP Growth Rate	4.2%	3.9%	3.1%	2.7%	2.3%	2.1%	2.9%
Consumer Price Inflation Rate	2.1%	2.0%	2.0%	2.0%	2.0%	2.0%	2.0%
Nominal Wage Growth Rate	3.6%	3.8%	3.9%	4.0%	3.8%	3.7%	3.8%
Treasury Bond Yield (3-year)	2.5%	2.2%	2.2%	2.1%	2.1%	2.1%	2.2%
GDP Deflator Inflation Rate	1.9%	2.0%	1.9%	1.9%	1.8%	1.8%	1.9%

Note: The average is the simple arithmetic mean of each variable during the forecast period (2025–2072). All units are in percent (%).

Source: National Assembly Budget Office (NABO), 2025.

²Statistics Korea's population projections (adjusted for military service) endogenously determine labor supply, participation, and unemployment, which converge to long-run equilibrium, while hours per worker stay constant. From 2025 onward, potential GDP is driven by exogenous TFP and demographic-driven labor input in a Cobb–Douglas production function, with 1970–2024 levels extracted via an HP filter. The saving rate is estimated using a regression on GDP growth and old-age dependency ratios, and inflation is forecast by a mid-term time series before settling at 2 percent long term (the GDP deflator is treated similarly). Interest rates follow a Taylor rule plus a time-varying term premium for the 3-year bond yield, and wage growth is tied to the marginal product of labor adjusted for expected CPI inflation.

2.3 Dim Fiscal Projection

As society ages, the rate of economic growth may decline and the social security deficit could rise, leading such economies to face the prospect of sustained deficits. Korea's central government is expected to experience significant deficits, primarily driven by a secular increase in social security spending resulting from its aging population.

Table 2: 2025 Long-Term Budget Outlook by NABO

Year, % of GDP	2025	2040	2050	2060	2072
Total Revenue	24.5%	24.7%	23.7%	22.3%	22.0%
Tax Revenue	14.3%	15.4%	15.4%	15.4%	15.3%
Non-tax Revenue	10.2%	9.3%	8.3%	6.9%	6.7%
Total Expenditure	25.5%	29.4%	30.9%	31.9%	33.6%
Mandatory Spending	13.9%	17.4%	18.9%	19.9%	21.6%
Social Welfare Spending	7.1%	9.7%	10.7%	11.3%	12.4%
Local Transfer Payments	5.3%	6.0%	6.0%	6.0%	6.0%
Interest Payments	1.1%	1.3%	1.8%	2.2%	2.8%
Other Mandatory Spending	0.4%	0.4%	0.4%	0.4%	0.4%
Discretionary Spending	11.6%	12.0%	12.0%	12.0%	12.0%
Social Security (SS) Surplus	2.3%	-0.2%	-2.0%	-3.9%	-5.2%
SS Operational Balance	0.4%	-1.9%	-3.1%	-4.0%	-5.2%
Total Deficits	1.0%	4.7%	7.2%	9.5%	11.6%
Total Deficits excl. Int. Payments	-0.1%	3.4%	5.4%	7.3%	8.8%
Total Deficits excl. SS and Int. Payments	2.2%	3.3%	3.1%	3.1%	3.2%
National Debt	47.8%	80.3%	107.7%	136.0%	173.0%

Note: Unit: % of GDP. The SS surplus includes investment income earned by the National Pension Fund, whereas the SS operational balance excludes it.

Source: Korean NABO Long-Term Fiscal Projections 2025–2072.

The Long-Term Fiscal Outlook released by the NABO provides a foundation for assessing the sustainability of public finances over the medium to long term. It is important to note that the outlook is based on the assumption that the existing system remains unchanged. Table 2 summarizes NABO's baseline projection under the medium-population variant. This scenario adopts Statistics Korea's middle-variant demographic path—implying a gradual population contraction from roughly 51.5 million in 2025 to about 36 million in 2072, only a modest recovery of the fertility rate to 1.08, and a steady rise in life expectancy—while keeping all current tax and expenditure rules unchanged.

The NABO projections reveal concerning fiscal dynamics characterized by a widening structural imbalance between revenues and expenditures. Total revenues are projected to decline from 24.5% of GDP in 2025 to 22.0% by 2072, while total expenditures are expected to rise substantially from 25.5% to 33.6% over the same period. This divergence results in persistent fiscal deficits,

expanding from 1.0% of GDP in 2025 to 11.6% by 2072.

The primary driver of deficit growth is the rapid deterioration of the social security surplus. Assuming current laws remain unchanged, Korea's rapidly aging society will inevitably lead to a substantial increase in the social security deficit. Social security is projected to run a surplus of 2.3% of GDP in 2025, but as the population ages it deteriorates steadily, breaking even around 2040 and turning to increasingly large deficits that reach about 5.2% of GDP by 2072.

Fiscal deterioration is further exacerbated by rising interest expenditures, which are projected to increase from 1.1% of GDP in 2025 to 2.8% by 2072. This reflects the compounding effects of mounting public debt and evolving interest rate dynamics. As a result, the consolidated deficit widens from 1.0% to 11.6% of GDP over the projection period. Sustained deficit spending drives government debt onto an explosive trajectory, rising from 47.8% of GDP in 2025 to 173.0% by 2072. Notably, this projection assumes a conservative estimate of debt servicing costs, with government bond yields held constant at around 2% throughout the period, despite the substantial deterioration in fiscal conditions. In contrast, other spending categories—including discretionary spending and transfers to local governments—remain relatively stable.

The fiscal concern extends beyond the social security deficit caused by an aging population. Even if we exclude social security and interest expenses from the total deficits, the government still runs consistent deficits of 2.2% to 3.3% of GDP (see the second-to-last row of Table 2). This indicates a concern for the fiscal discipline of the Korean government, even when social security is excluded. This also suggests that the current surplus in the social security system actually masks the problem of excessive government spending, making the total deficit appear smaller. This pattern underscores that the threat to fiscal sustainability stems not only from future demographic pressures and entitlement programs but also from ongoing excessive government spending.

3 The Consolidated Public Sector Balance Sheet

To understand how prepared the Korean government is to face future fiscal challenges, we analyze the consolidated balance sheet of the public sector, following the methodology of [Chien et al. \(2025a\)](#), which was originally applied to Japan. To the best of our knowledge, this study is the first to evaluate the sustainability of Korea's fiscal position through the lens of its consolidated public sector balance sheet.

In the literature on fiscal sustainability, the majority of studies focus solely on central government debt when evaluating government solvency. This approach may be appropriate for countries such as the United States, where the government holds relatively few financial assets and most liabilities take the form of public debt. However, we argue that focusing exclusively on national debt may provide a biased view of Korea's public finance situation. Hence, consolidating the balance sheets across different parts of the public sector can provide a more comprehensive view of the government's fiscal position.

3.1 Why Consolidated Balance Sheet?

We examine the consolidated balance sheet for at least three reasons. First, any profits or losses incurred by publicly owned institutions — central and local governments, social security funds, sovereign wealth funds, and the central bank — will ultimately be borne by the general public. The institutional channels are concrete: the Bank of Korea remits its net profit to the government under Article 99 of the Bank of Korea Act (after allocating legal reserves); the KIC invests assets entrusted by the Ministry of Economy and Finance and the Bank of Korea, so its investment gains and losses accrue to those entrusting institutions; and the National Pension Fund is a public reserve whose performance directly affects the government’s implicit pension obligations. There is therefore little justification for focusing solely on the central government while ignoring these other public institutions.

Second, these public institutions often hold sizable asset positions, particularly in social security funds or sovereign wealth funds. This is certainly the case in Korea, where the current asset position of the public pension fund is substantial — around 46% of GDP in 2024. Similarly, the sovereign wealth fund managed by the Korea Investment Corporation (KIC) invests globally and represents over 11% of Korea’s GDP. Both the central government and the Bank of Korea invest abroad through KIC. To accurately assess the government’s indebtedness, we should consider not only net debt (by offsetting liabilities with assets) instead of gross debt but also the composition of its balance sheet.

Third, consolidation is the theoretically appropriate level at which to analyze fiscal sustainability. At least since [Sargent and Wallace \(1981\)](#), it has been well understood that the central bank and the general government face the same intertemporal budget constraint, so that the consolidated public sector—rather than any single institution—is the natural object of analysis. A recent literature revisits this consolidated budget constraint under risk, including [Chien et al. \(2025a\)](#) and [Jiang et al. \(2024\)](#); in a similar spirit, [Hall and Sargent \(2022\)](#) organize their fiscal accounting around a consolidated government budget constraint. Consolidating in this way naturally brings the asset side of the public sector—not just its outstanding debt—into the analysis, and it is this consolidated approach that we adopt here.

In short, the consolidated balance sheet provides a more comprehensive view of the sustainability of public finances. Adopting this consolidated approach broadens the scope of public finance policies beyond fiscal policy to include exchange rate and interest rate policies. This perspective helps identify potential policy risks and challenges, as well as viable policy options.

3.2 Increasing Risk Exposure of Public Sector

We consolidate the balance sheets of public sector, which includes the central government, local governments, the social security fund, the Bank of Korea (BoK) and the Korean Investment Corporation (KIC). The data is available from the Korea Flow of Funds, maintained by the BoK. Table

3 provides a snapshot of the public sector's balance sheet in 2008, 2015 and 2024. These dates are chosen to represent the start, the middle and the end of our dataset. All numbers are expressed as a percentage of GDP.

Our consolidation is organized around the entities whose financial gains and losses ultimately accrue to taxpayers: the central and local governments, the National Pension Service (NPS), the KIC, and the Bank of Korea. Relative to Korea's official debt aggregates, this perimeter lies between *D2* and *D3*—it covers the general government (broadly *D2*) and adds the Bank of Korea and the KIC, but excludes the non-financial public corporations that make up the remainder of *D3*. On the asset side we make two deliberate exclusions: non-financial assets such as public real estate and infrastructure, which are illiquid, hard to value, and not available to service debt or pensions; and pay-as-you-go programs such as the National Health Insurance and long-term care insurance, which hold no investment fund to bring onto the balance sheet.³

Constructing this consolidated balance sheet yields Table 3, which reports the public sector's assets and liabilities for 2008, 2015, and 2024. Reading the 2024 column, the public sector held financial assets worth 117.1% of GDP against liabilities of 71.2% of GDP, leaving a positive net asset position of 45.9% of GDP. The two sides differ sharply in their risk profile. On the asset side, about 73% of the holdings are risky: a long position of 53.3% of GDP in foreign assets (exposing the public sector to currency risk) and a further 31.7% of GDP in domestic equities, fund shares, and non-government debt securities. The liability side, in contrast, is almost entirely safe and short in duration—net government bonds of 37% of GDP (44.9% gross, net of the public sector's own holdings), together with monetary-policy instruments such as currency (7.4%), bank reserves (4.4%), and central bank securities (4.5% of GDP).

Comparing the two endpoints shows how this position was built up. In 2008 the balance sheet was much smaller, with financial assets of 88.9% of GDP and foreign assets of only 23.3% of GDP, most of which were foreign reserves. Total assets then rose by 28.2%p of GDP between 2008 and 2024, an increase driven entirely by risky foreign assets: these grew by 30.0%p of GDP—more than the total rise, as domestic categories edged down—with foreign equities alone climbing by 21.2%p (from 1.1% to 22.3% of GDP). Liabilities rose by a more modest 14.4%p of GDP, almost all of it safe domestic government debt, as government bonds increased by 15.7%p of GDP (partly offset by a decline in central bank securities).

Putting the two sides together, the public sector's net asset position climbed from 32.1% of GDP in 2008 to 45.9% in 2024, a gain of 13.8%p. On a net-worth basis, then, the consolidated public sector appears healthier at the end of the sample than at its start—which makes the *composition* of the balance sheet, rather than its net size, the central issue.

³Health and long-term care insurance are nonetheless likely to become the more pressing source of demographic fiscal pressure over the medium term, especially as the 2025 pension reform pushes the projected depletion of the National Pension Fund roughly four decades out. This does not weaken our conclusion: our deficit and debt paths rest on NABO's general-government projections, which already embed the rising, aging-driven health and welfare spending, and consolidating these programs as separate funds would only enlarge the projected shortfall.

Table 3: Consolidated Balance Sheet of Public Sector

% of GDP, Year End	2008	2015	2024	08-24 Diff
Assets				
Deposits (Net)	6.0%	3.2%	4.8%	-1.2%p
Government Loans	13.5%	10.7%	11.0%	-2.5%p
BOK Loans	1.9%	1.1%	1.1%	-0.8%p
Debt Securities	17.1%	17.2%	15.1%	-2.0%p
<i>Government Bonds</i>	9.6%	8.6%	7.7%	-1.9%p
<i>Other Debt Securities</i>	7.5%	8.6%	7.3%	-0.2%p
Equities and Shares	19.5%	23.3%	24.4%	4.9%p
Foreign Assets	23.3%	33.4%	53.3%	30.0%p
<i>Foreign Debt Securities</i>	0.7%	0.6%	1.9%	1.2%p
<i>Foreign Equities</i>	1.1%	6.8%	22.3%	21.2%p
<i>Other Foreign Claims</i>	1.0%	3.3%	7.0%	6.0%p
<i>Foreign Direct Investment</i>	0.0%	1.0%	5.3%	5.3%p
<i>BoK Foreign Claims</i>	20.6%	21.7%	16.9%	-3.7%p
Others	7.6%	7.0%	7.3%	-0.3%p
Sum (Assets)	88.9%	95.9%	117.1%	28.2%p
Liabilities				
Currency	2.6%	4.9%	7.4%	4.8%p
Bank Reserves	4.5%	3.7%	4.4%	-0.1%p
Central Bank Securities	10.8%	10.5%	4.5%	-6.3%p
Government Loans	1.3%	0.8%	1.0%	-0.3%p
Government Bonds	29.2%	39.1%	44.9%	15.7%p
Domestic Equities	1.7%	1.7%	2.0%	0.3%p
Foreign Assets	0.8%	0.5%	0.6%	-0.2%p
Others	5.9%	6.3%	6.4%	0.5%p
Sum (Liabilities)	56.8%	67.6%	71.2%	14.4%p
Net Assets	32.1%p	28.3%p	45.9%p	13.8%p

Note: Unit: % of GDP.

Source: Korea Flow of Funds.

This rising net worth, however, masks a growing concentration of risk: the expansion of the asset side is overwhelmingly an expansion of *risky* assets, and it is concentrated in two institutions. Tables 4 and 5 report the asset holdings of the NPS and the KIC. The social security fund managed by the NPS reached about 46.5% of GDP in 2024, with more than half invested abroad—its large foreign position effectively makes it a de facto sovereign wealth fund. The KIC, to which both the central government and the Bank of Korea entrust foreign assets, grew from 1.8% to 11.7% of GDP over 2008–2024, mostly through foreign equities and bonds.⁴ A comparison of Tables 3,

⁴These entrusted assets are reported in the Flow of Funds under various categories for both the BoK and the general government; for instance, assets entrusted by the BoK appear under “Bank of Korea – Other External Debt” and those entrusted by the central government under “General Government – Other External Debt” (or, for ineligible portions, by asset type such as “Non-resident Issued Equities” or “Overseas Bonds”). We therefore recover the amounts invested through the KIC from its annual reports.

4, and 5 makes clear that the public sector’s growing risky-asset exposure is driven by the asset accumulation and portfolio shifts of the NPS and the KIC—the engine of the carry trade at the center of this paper.

Table 4: Social Security Fund Asset Holdings

% of GDP, Year End	2008	2015	2024	08 - 24 Diff
Domestic Assets	17.5%	21.0%	19.6%	2.1%p
Debt Securities	15.1%	15.5%	13.2%	-1.9%p
Equities	2.4%	5.5%	5.4%	3.0%p
Alternatives	0.0%	0.0%	1.0%	1.0%p
Foreign Assets	1.2%	5.3%	27.0%	25.8%p
Debt Securities	0.8%	1.3%	3.4%	2.6%p
Equities	0.5%	4.0%	16.6%	16.1%p
Alternatives	0.0%	0.0%	7.0%	7.0%p
Total Assets	18.7%	26.3%	46.5%	27.8%p

Note: Unit: % of GDP.

Source: NPS annual Report.

Table 5: Asset Holdings of KIC

% of GDP, Year End	2008	2015	2024	08 - 24 Diff
Foreign Debt Securities	1.3%	2.1%	3.7%	2.4%p
Foreign Equities	0.5%	2.4%	4.6%	4.1%p
Foreign Alternative	0.0%	0.3%	2.6%	2.6%p
Others	0.1%	0.7%	0.8%	0.7%p
Sum	1.8%	6.0%	11.7%	9.9%p

Note: Unit: % of GDP.

Source: KIC annual Report.

The Korean government’s rapid shift toward riskier investments has enhanced returns on its public pension fund and sovereign wealth fund by increasing exposure to financial markets. We emphasize that this consolidated carry trade need not reflect a coordinated or intentional policy at the level of the public sector as a whole; rather, it emerges as the aggregate consequence of the separate investment mandates of the NPS and the KIC, which independently seek higher returns, combined with the central government’s reliance on low-cost domestic debt financing. Whether deliberate or not, the resulting position appears, in effect, to anticipate the growing fiscal burden associated with the country’s rapidly aging population.

3.3 The Carry Trade Return

To estimate the public sector’s balance sheet return, we rely heavily on the annual reports of the NPS and KIC. These reports provide detailed information on asset categories, portfolio composi-

tion, realized returns and benchmark returns. We use this information to augment the portfolio composition of the public sector balance sheet by incorporating the investment choices made by both institutions.

Assumptions. To assign returns to each balance-sheet category, we make several key assumptions. First, the amount of debt securities held by the public sector roughly corresponds to the holdings reported by the NPS. We therefore assume that the return on public sector debt securities follows the return on the NPS’s domestic debt securities category. Specifically, we use the return on the KIS Corporate Bond Index as a proxy for the return on government’s holdings of debt securities, covering both NPS and non-NPS portions. Second, the combined holdings of domestic equities and investment fund shares by the public sector far exceed the total of domestic equities and alternative investments reported by the NPS. We then divide the public sector’s equity holdings into two parts: the portion attributed to the NPS, and the residual, which accounts for other public institutions’ holdings. The non-NPS holding of equity is quite sizeable, almost 18% of GDP in 2024. In addition to NPS, there are also other government funds that invest in equity-related financial products. Examples include the Housing and Urban Development Fund, Credit Guarantee Fund, and Science and Technology Promotion Fund.⁵ We simply assume that the return on all equity holdings (NPS and non-NPS combined) follows that of the Korea Composite Stock Price Index (KOSPI). Finally, the foreign asset holdings of the public sector are consistently larger than the combined foreign assets of the NPS and KIC. This discrepancy arises because only part of the Bank of Korea’s (BoK) foreign reserves is entrusted to the KIC, and the exact amount delegated is not publicly disclosed. As a result, we divide foreign assets into two components: the portion managed by the NPS and KIC, and the residual, which represents the foreign reserves managed internally by the BoK. This decomposition also explains why the *increase* in the combined NPS and KIC foreign assets between 2008 and 2024 exceeds the increase in the consolidated foreign-asset total: the BoK’s own foreign reserves declined over the period (from 20.6% to 16.9% of GDP), partially offsetting the sharp rise in NPS and KIC foreign holdings. Under these assumptions, the modified government balance sheet is reported in Table 6.

Data. We then utilize the return information reported by the NPS and KIC to assign returns to these foreign assets. In particular, the returns on the Global Equity category are benchmarked against the returns on the MSCI ACWI Net Total Return Index excluding Korea. Both the Global-Fixed-Income and the Global-Alternatives-plus-Others categories utilize customized indices developed by the NPS. Appendix D demonstrates that our return proxies track the realized annual returns of NPS and KIC well.

⁵This non-NPS residual (18.1% of GDP in 2024) is constructed as the Flow of Funds total for the public sector’s holdings of investment-fund shares and domestic equities, net of the NPS’s own reported domestic equity and fund holdings (6.3% of GDP). It reflects the idle balances of public funds that are placed, directly or through investment-fund shares, in equity-linked products rather than held as deposits.

Table 6: Modification of Public Sector's Balance Sheet

% of GDP, Year End	2008	2015	2024	08-24 Diff
Assets				
Deposits (Net)	6.0%	3.2%	4.8%	-1.2%p
Government Loans	13.5%	10.7%	11.0%	-2.5%p
BOK Loans	1.9%	1.1%	1.1%	-0.8%p
Domestic Debt Securities	17.1%	17.2%	15.1%	-2.0%p
NPS	15.2%	15.2%	13.2%	-2.0%p
Non-NPS	1.9%	2.0%	1.9%	0.0%p
Equities and Fund Shares	19.5%	23.3%	24.4%	4.9%p
NPS	3.0%	6.6%	6.3%	3.3%p
Non-NPS	16.5%	16.6%	18.1%	1.6%p
Foreign Assets	23.3%	33.4%	53.3%	30.0%p
Global Equity	1.4%	6.4%	21.2%	19.8%p
NPS	0.5%	4.0%	16.6%	16.1%p
Non-NPS	0.9%	2.4%	4.6%	3.7%p
Global Fixed Income	2.9%	3.3%	7.1%	4.2%p
NPS	0.8%	1.3%	3.4%	2.6%p
Non-NPS	2.1%	2.0%	3.7%	1.6%p
Global Alternatives + Others	0.3%	3.4%	10.4%	10.1%p
NPS	0.0%	0.0%	7.0%	7.0%p
Non-NPS	0.3%	3.4%	3.4%	3.1%p
FX Reserves	18.8%	20.3%	14.6%	-4.2%p
Others	7.6%	7.0%	7.3%	-0.3%p
Sum (Assets)	88.9%	95.9%	117.1%	28.2%p
Liabilities				
Currency	2.6%	4.9%	7.4%	4.8%p
Bank Reserves	4.5%	3.7%	4.4%	-0.1%p
Central Bank Securities	10.8%	10.5%	4.5%	-6.3%p
Government Loans	1.3%	0.8%	1.0%	-0.3%p
Government Bonds	29.2%	39.1%	44.9%	15.7%p
Domestic Equities	1.7%	1.7%	2.0%	0.3%p
Foreign Assets	0.8%	0.5%	0.6%	-0.2%p
Others	5.9%	6.3%	6.4%	0.5%p
Sum (Liabilities)	56.8%	67.6%	71.2%	14.4%p
Net Assets	32.1%	28.3%	45.9%	13.8%p

Note: Unit: % of GDP.

Source: Korea Flow of Funds, Annual Reports of NPS and KIC.

We proxy the return on Korea's foreign reserves using the ICE BofA 5–7 Year U.S. Treasury Index, translated into won at the year-end exchange rate as for every other foreign asset. Detailed portfolio data on Korea's foreign-exchange reserves are not publicly available, so we assume that a significant portion is invested in U.S. Treasury securities. Although a U.S. Treasury report puts the average maturity of the *long-term* U.S. debt held by foreign official institutions at about 8.5 years over 2015–2022 (see [Foreign Portfolio Holdings of U.S. Securities, 2023](#), Page 24), reserves also

hold substantial shorter-dated instruments and are managed for liquidity; we therefore adopt a short-to-intermediate (5–7 year) duration rather than a long-duration (10+ year) proxy, which—as discussed below—would overstate the reserves’ return volatility.

Table 7: Return or Index Used for Asset and Liability Items

Item	Proxy Variable / Index	Source
Assets		
Deposits (Net)	Liquidity Adjustment Deposits Rate	BoK
Government Loans	Public Funds Management Fund (PFMF)	MOEF
BOK Loans	Standing Lending Facility	BoK
Domestic Debt Securities		
NPS	KIS Corporate Bond Index	Bloomberg
Non-NPS	KIS Corporate Bond Index	Bloomberg
Equities and Fund Shares		
NPS	KOSPI Index	Bloomberg
Non-NPS	KOSPI Index	Bloomberg
Foreign Assets		
Global Equity	MSCI ACWI index Net Total Return (ex Korea)	Bloomberg
Global Fixed Income	NPS Customized Index	Authors
Global Alternatives + Others	Customized Index	Authors
FX Reserves	ICE BofA 5–7 Year US Treasury Index	Bloomberg
Liabilities		
Currency	Zero return	—
Bank Reserves	Bank reserve rate	BoK
Central Bank Securities	MSB 91 days	BoK
Government Loans	Public Funds Management Fund (PFMF)	MOEF
Government Bonds	KIS Government Bond Index	Bloomberg
Domestic Equities	KOSPI Index	Bloomberg
Foreign Assets	Credit Suisse Hedge Fund index	Bloomberg

Note: Source abbreviations—BoK: Bank of Korea; MOEF: Ministry of Economy and Finance. Market indices (KOSPI, MSCI ACWI, ICE BofA, Credit Suisse, KIS) are obtained from Bloomberg; “Authors” denotes series we construct from the National Pension Service’s published benchmark weights. The NPS Customized Index for Global Fixed Income follows the NPS benchmark allocation of 50% Treasuries, 20% government-related securities, and 30% corporate bonds, all hedged to the Korean won; the Customized Index for Global Alternatives + Others assigns weights of 25% to foreign equity returns and 75% to foreign bond returns to approximate the NPS’s global alternatives portfolio.

In addition to the assets held by the NPS and KIC, the public sector holds other substantial assets. One notable category is government loans, which amounted to approximately 11% of GDP in 2024. These include policy loans such as those to small and medium-sized enterprises (SMEs), farmland funds, and student loans. The return on these policy loans is then approximated by the lending rate on the Loan Account of the Public Funds Management Fund (PFMF). The PFMF is an umbrella fund through which the Korean government channels policy loans—for example, to SMEs, housing programs, and infrastructure projects. The Loan Account records the interest rates actually charged by the government on these loans.⁶

⁶The Public Funds Management Fund (PFMF) is an umbrella account through which the Korean government chan-

For the foreign assets category, the return on Global Fixed Income is based on the NPS’s customized benchmark, which consists of 50% Treasuries, 20% government-related securities, and 30% corporate bonds, all hedged to the Korean won. This specification directly follows the benchmark setting used in the NPS Fund for its global fixed income portfolio. For the Global Alternatives + Others category, the return is constructed by assigning weights of 25% and 75% to foreign equity and foreign bond returns, respectively. These weights are chosen to closely replicate the historical return profile of the NPS’s global alternatives portfolio, which is approximated using a combination of foreign equity and foreign bond returns.

Two remaining asset types are deposits and BOK loans, both of which are linked to policy rates. We use the Liquidity Adjustment Deposit Rate and the Standing Lending Facility Rate, published by the Bank of Korea, to proxy their respective returns.

On the liabilities side, the portfolio returns are relatively straightforward to estimate. The return on currency is set to be zero. For liabilities associated with monetary and exchange rate policy—specifically, bank reserves and central bank securities—their returns are linked to the bank reserve rate and the 91-day Monetary Stabilization Bond rate, respectively. In the Korean context, the bank reserve rate is effectively set at 0%, reflecting the fact that the Bank of Korea does not pay interest on required reserves. We use the KIS Government Bond Index to calculate the return on government bonds. The remaining liabilities include government loans, domestic equity, and foreign assets, with returns proxied by the PFMF rate, the KOSPI Index, and the Credit Suisse Hedge Fund Index, respectively. In calculating the returns on the consolidated public sector balance sheet, we exclude residual items categorized as “Others” on both the asset and liability sides. As noted in Table 6, the shares of “Others” are relatively small and similar in magnitude across both sides of the balance sheet—approximately 7.3% of GDP for assets and 6.4% for liabilities in 2024. Given these near-offsetting positions and their overall small weights, including them would have minimal impact on the net return calculation. Thus, omitting them simplifies the analysis without materially affecting the results. We include the Bank of Korea’s foreign-exchange reserves in the spread calculation, but proxy their return with a short-to-intermediate (5–7 year) US Treasury index rather than a long-duration (10+ year) one, translated at the year-end exchange rate as for every other foreign asset. This choice reflects the liquidity-oriented, shorter-duration composition of the reserves and avoids overstating their volatility: a long-duration proxy generates implausibly large valuation swings—most starkly a one-off gain during the 2008 won depreciation—that reflect mark-to-market interest-rate sensitivity rather than the reserves’ underlying investment performance.

The returns for each asset and liability category on the consolidated balance sheet are calculated various policy loans—for example, to small and medium-sized enterprises (SMEs), housing programs, infrastructure projects, and social services. The Loan Account of the PFMF records the lending rates actually charged by the government on these loans. Official quarterly lending rate announcements, such as those for 2025Q3, are published by the Ministry of Economy and Finance and are available at <https://www.moef.go.kr>. In our analysis, these lending rates are used as the return measure for the “Government Loans” category in the consolidated public sector balance sheet.

lated on an annual basis. Table 7 summarizes all the returns or indices used for the consolidated balance sheet.

Results. Combining the estimated return series with the asset and liability holdings reported in the two consolidated balance-sheet tables, we compute the return earned on each side of the balance sheet and the spread between them, summarized in Figure 3. Panel (a) plots the asset return and the liability cost, both in percent per year; their difference—the asset return minus the liability cost—is the return spread. Panel (b) reports the resulting net investment income as a share of GDP (bars) together with that spread (line). Read as period averages, the figure shows that over the full sample (2008–2024) the public sector earned an asset return of 5.2 percent against a liability cost of 3.1 percent—a spread of 2.2 percentage points a year, equivalent to a net return of about 3.2 percent of GDP annually—while over the more recent 2015–2024 window the spread is larger still, at 3.5 percentage points.

This spread has widened markedly since 2015. In the earlier period (2008–2015) it was only marginally positive, at 0.2 percentage points, so that the modest average return of 1.5 percent of GDP came mainly from the public sector’s positive net asset position rather than from any excess return; over 2015–2024, by contrast, the spread rose to 3.5 percentage points and the average annual return to 4.3 percent of GDP. The widening reflects the portfolio’s growing tilt toward risky assets. As documented above, the asset side expanded chiefly in foreign equities and other higher-yielding securities, whereas the liability side remained concentrated in safe, low-yielding instruments; because the expanding risky assets earn more than the safe liabilities cost, the gap between the two has grown. In short, it is the rising share of risky assets that drives the widening spread.

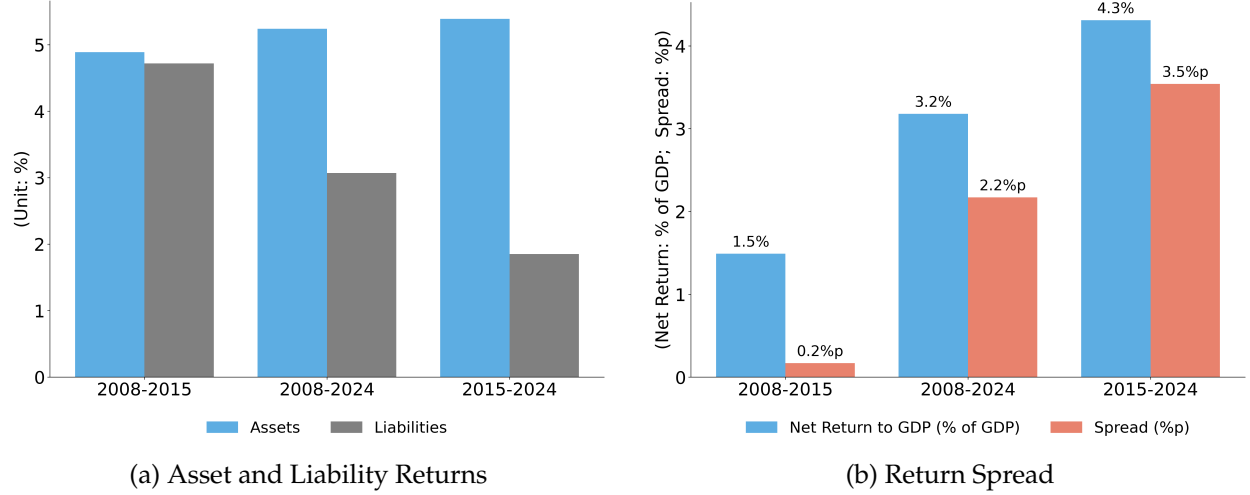
Taken together, these facts describe a *carry trade*. A carry trade finances a position in higher-yielding, riskier assets by borrowing through a lower-yielding, safer instrument, earning the spread between the two. That is precisely what the consolidated public sector does: it funds itself with safe, low-yielding domestic liabilities—government bonds and monetary-policy instruments—and invests the proceeds in risky, higher-yielding assets, predominantly the foreign equities and other securities held through the NPS and the KIC. The average spread of 2.2 percentage points, rising to 3.5 since 2015, is precisely the carry that this position earns.

These balance sheet gains are consistent with actual fiscal outcomes. Although the central government persistently ran deficits, the consolidated public sector’s net asset position nonetheless improved, which suggests that the excess return earned on the balance sheet, together with the surpluses of the social security funds, played a crucial role in buffering the government’s fiscal burden. The excess returns generated by these risky investments are therefore a key factor that any assessment of Korea’s fiscal sustainability must take into account.

In sum, the carry trade has kept the public sector’s net worth not only positive but rising, even as the central government ran persistent deficits. That very success, however, raises the central

question of this paper: at what price does it come, and can it be sustained? The next section takes up that question.

Figure 3: Estimated Returns on the Consolidated Public Sector Balance Sheet



Note: Panel (a) plots the asset return and the liability cost (percent per year); the return spread is their difference, asset return minus liability cost, in percentage points (%p). In panel (b), the bars show net investment income as a share of GDP (percent of GDP, %) and the line shows the return spread (%p). The average spread is 0.2%p over 2008–2015 and 3.5%p over 2015–2024, for a full-sample average of 2.2%p.

4 Sustainability of Korea Public Finance

We proceed in three steps. First, we evaluate the government’s carry-trade strategy itself and ask who ultimately bears the risk it creates. Second, we assess fiscal sustainability from a present-value (NPV) perspective, comparing the net present value of future deficits with the public sector’s net asset position. Third, we examine sustainability dynamically through a VAR-based simulation of the consolidated balance sheet.

4.1 Who Bears the Risk

Our sustainability analysis begins by assessing the government budget constraint. Let G_t denote nominal government spending before interest expenses on the debt, T_t denote nominal government tax revenue, and $S_t = T_t - G_t$ denote the nominal primary surplus. We use P_t^S to denote the cum-dividend price of the surplus claim. If the transversality condition is satisfied, then the value of debt is fully backed by the present value of future surpluses and the value of government assets: $P_t^S + A_t = D_t$, which represents the government budget constraint in present value form. Here D_t denotes the market value of outstanding public debt—equivalently, the financing requirement that, under the constraint, must be covered by the present value of future primary surpluses P_t^S .

and the value of government assets A_t . Consistent with our consolidated balance-sheet scope, A_t comprises the public sector's financial assets; non-financial assets such as public real estate are excluded, as they are illiquid, difficult to value, and not readily available to service debt or pension obligations (their inclusion would not materially alter the conclusions).

The government's budget constraint also disciplines the returns it earns on its balance sheet. Let R_t^A , R_t^D , and R_t^S denote the returns on government-held assets, on government debt, and on the surplus claim between periods $t - 1$ and t , and let R_t^f denote the risk-free return. As shown in Appendix A, a few algebraic steps turn the budget constraint into the risk-premium equivalence relation

$$\underbrace{\mathbb{E}_t [R_{t+1}^A - R_t^f]}_{\text{risk premium on public assets}} = \frac{D_t}{A_t} \underbrace{\mathbb{E}_t [R_{t+1}^D - R_t^f]}_{\text{borne by bondholders}} + \left(1 - \frac{D_t}{A_t}\right) \underbrace{\mathbb{E}_t [R_{t+1}^S - R_t^f]}_{\text{borne by taxpayers}}, \quad (1)$$

which states that the expected excess return on government assets is a weighted average of the excess returns on its debt and on the surplus claim. Its central message is that the government cannot eliminate the risk of its investments; it can only redistribute it—to bondholders or to taxpayers (Jiang et al., 2024).

Korea's rapidly aging population places its intertemporal budget constraint under mounting strain. Rather than pursuing politically sensitive measures such as benefit cuts or tax increases, the Korean government has adopted an alternative strategy aimed at improving the return on public assets relative to liabilities. Section 3 shows that this has involved repositioning the national pension fund to function more like a sovereign wealth fund, increasing its exposure to higher-risk, higher-return investments. This shift represents an effort to close the fiscal gap by generating additional returns on public wealth, thereby reducing the need for immediate fiscal consolidation. Over the past decade, this strategy appears to have yielded tangible benefits, as the government's net asset position has strengthened despite running primary fiscal deficits.

Could this risky investment strategy help the Korean government sustain its future fiscal burden? And what risks does the public sector assume in the process? To address these questions, we therefore analyze the risk exposure that the government assumes.

The equivalence in equation (1) leaves the government two options, depending on where it lets this risk land. Consider each in turn.

Consider a scenario in which the government seeks to protect future taxpayers by shifting risks to bondholders. In this case, bondholders would demand a higher risk premium, as indicated by equation (1). As a result, the government would not be able to earn a higher average return relative to its funding cost, since the increased borrowing cost would fully offset any additional returns from risky investments. In other words, the risk-taking approach implemented by the Korean government cannot alleviate its fiscal burden, as the gains from risky assets are neutralized by the higher cost of debt.

Consider the alternative case in which the government maintains its risk-free commitments to bondholders. In this scenario, the risk is instead borne by future taxpayers. The excess return earned from the government's risky investments represents compensation for the risks ultimately absorbed by taxpayers. However, this approach implies higher taxation during economic downturns, precisely when asset returns are low. Imposing greater fiscal burdens during recessions is not only politically unpopular but also potentially unfeasible.

This risk-taking fiscal strategy is rooted in a fundamental risk mismatch: the government makes risk-free promises to both pensioners and bondholders, yet these promises are backed by inherently risky sources of income—namely, tax revenues and returns on investments that fluctuate with economic cycles.

Korea's fiscal sustainability, therefore, depends heavily on this unresolved risk mismatch. While the mismatch has remained largely hidden thus far—thanks to the government's remaining fiscal capacity and the absence of major investment losses—it poses a growing vulnerability. As the burden of social security rises, the government is likely to exhaust its fiscal capacity, and hence fiscal tension could become more apparent. In the event of a significant economic downturn, risky government investments could suffer substantial losses, and raising taxes may become both economically damaging and politically unviable. At that point, the government may face a difficult choice between protecting taxpayers or honoring its commitments to bondholders. It is far from certain that the government would uphold its risk-free promises to bondholders under such conditions. We now assess quantitatively whether this strategy can nonetheless sustain Korea's public finances, beginning with a present-value accounting of future deficits.

4.2 The NPV of Future Deficits

In the theoretical literature on public finance, when the government faces a deterministic path of future spending, such spending should be financed through taxation rather than debt. If taxes are distortionary, the tax-smoothing principle applies—meaning the tax burden should be distributed evenly over time. For stochastic or unexpected government spending, the government may use debt finance in order to smooth the tax burden over time (Barro, 1979; Lucas and Stokey, 1983).

Korea's major fiscal challenge is the secular rise in social security spending caused by population aging. Because this future spending is largely predictable, the tax-smoothing logic above implies that it should be prepared for in advance rather than financed by debt as it materializes. The natural question is then whether the resources the public sector has already accumulated are sufficient to meet it. Under the government budget constraint, this amounts to asking whether the current net asset position can cover the present value of the future deficits that population aging will generate—a question we take up in this subsection through a net-present-value (NPV) analysis.

We begin with the net asset position itself. As reported in Table 3, the public sector's total financial assets and liabilities in 2024 were 117% and 71.2% of GDP, leaving a net asset position of

45.9% of GDP. Under the government budget constraint, this is the largest present value of future deficits that the public sector can afford to run.

To make this comparison precise, let $M_{t,t+s}$ be the real stochastic discount factor between time t and $t + s$. The net present value of the surplus at time t is

$$\text{NPV}_t^S = \sum_{s=0}^{\infty} \mathbb{E}_t [M_{t,t+s} S_{t+s}] \equiv \sum_{s=0}^{\infty} \frac{\mathbb{E}_t S_{t+s}}{R_{t,t+s}^S} \quad (2)$$

where $R_{t,t+s}^S$ is the risk-adjusted discount rate for surplus claims. The projection is sustainable if the present value of future deficits, $-\text{NPV}_t^S$, is no larger than the net asset position of 45.9% of GDP; if it is larger, the government must eventually raise taxes, cut spending, or increase the value of its assets to satisfy the budget constraint. We therefore treat the fiscal situation as unsustainable when the NPV of future deficits far exceeds the net asset position.

Primary deficits consist of several components that differ in their exposure to aggregate risk, and because a riskier claim must be discounted at a higher rate, each warrants its own risk-adjusted discount rate. We assign these rates to the non-social-security and the social-security components in turn.

We apply a risk-adjusted discount rate to government tax revenues and expenditures other than social security benefits. Government surpluses tend to be pro-cyclical, which makes them riskier than a claim on GDP. Under the conservative assumption that central government revenues and expenditures are constant fractions of output, surpluses inherit the same risk profile as GDP—making them as risky as an unlevered equity claim on aggregate output.

In contrast, social security promises may be safer than a claim on GDP, as payments to current pensioners do not vary with economic conditions. However, benefits to future pensioners are calculated based on their lifetime contributions, which depend on GDP growth. As a result, these pension promises are less risky than a claim on GDP but are not effectively risk-free. We then calculate two net present values of social security benefits using both a risk-adjusted and a risk-free discount rate, serving as the lower and upper bounds of its NPV estimate.

We approximate the long-term nominal risk-free rate using the Korean 10-year government bond yield, which averaged around 2.4% between 2015 and 2024.⁷ The average inflation rate during the same period was 1.87%, which together with 2.4% bond yield imply a real long term risk free rate at 0.53%.⁸ We acknowledge that this risk-free status is itself state-contingent: if fiscal conditions were to deteriorate sharply, the sovereign yield would embed a default premium and could no longer be treated as risk-free. Our conclusions are nonetheless robust to reasonable variation in this rate, because a higher risk-free rate lowers the present value of *both* the safe pension and debt promises and the risky revenues that back them; the central finding—that the present value of future deficits far exceeds the net asset position—is unchanged. The long-run

⁷See the FRED data [10-Year Korea Government Bond Yields](#)

⁸The data is available here: [Korea Inflation Rate](#)

average nominal return of the KOSPI (Korea Composite Stock Price Index) is about 8%, implying an equity risk premium of 5.6%. According to the Financial Soundness Indicators published by the IMF, the average debt-to-equity ratio for Korean corporations was 1.35 over the past five years, which implies an unleveraged beta (asset beta) of 0.425. The implied GDP risk premium is then calculated as $5.6\% \times 0.425 = 2.38\%$.

With the discount rates in hand, we take the deficit projections from NABO's 2025 long-term outlook, in which the central government runs primary deficits through the end of the projection horizon in 2072, while social security runs a surplus until 2040 before turning to a deficit from 2041 through 2072. Two assumptions frame the calculation. First, we ignore the Bank of Korea's profits: under our consolidated balance-sheet approach these seigniorage revenues should count as government revenue, but they are small—about 2.5 trillion Korean won, or roughly 0.1% of GDP, in 2022—so omitting them is immaterial. Second, even though the projected deficit is still large and rising in 2072, we assume the government begins running a balanced budget from 2073. This is an oversimplified but deliberately conservative assumption: because the projected deficit is still large and growing, incorporating the post-2072 deficits would only widen the present-value shortfall. Our estimate should therefore be read as a lower bound on the true fiscal burden, not as a claim that the budget actually balances.

Under our assumptions, we calculate the present value of future deficits from the central government and the social security fund—here the National Pension Service (NPS), which dominates Korea's public pension system. For the central government alone, the net present value (NPV) of deficits up to 2072 amounts to 100% of 2024 GDP. In other words, the central government cannot sustain its own spending plan even if the social security is excluded:

$$45.9\% \text{ of GDP} \ll NPV_{2024-2072}^{\text{Deficits, Govt}} = 100\% \text{ of GDP},$$

which dwarfs the current net asset position of Korea's public sector, 45.9% of GDP.

The bleak projection of the social security surplus clearly exacerbates fiscal concerns. The net present value (NPV) of social security (NPS) deficits, calculated using a risk-free discount rate on expenditure and risk-adjusted rate in revenue, is high — amounting to 215% of GDP. This large present value stems from the asymmetric discounting applied to revenues and expenditures. Under this approach, the government effectively makes risk-free promises to current and future pensioners, while those promises are backed by risky social security tax revenues. As a result, even if the projected revenue and expenditure streams are identical, the present value of the deficit is not zero. As noted earlier, pension benefits are less risky than a claim to GDP but not entirely risk-free, since GDP growth affects payouts to future pensioners. This high NPV serves as an upper bound.

Alternatively, we calculate the NPV of social security deficits using the same risk-adjusted discount factors for both revenues and expenditures. Under this scenario, the NPV of projected

social security deficits drops to 64% of GDP, which can be regarded as a lower bound. This exercise implies that if pension benefits (for both current and future retirees) are indexed to GDP growth—which inherently carries aggregate risk—then this lower-bound estimate becomes valid, as the pension benefits inherit a similar level of risk as a GDP claim.

Taken together, the present value of social security deficits lies between a lower bound of 64% and an upper bound of 215% of GDP:

$$45.9\% \text{ of GDP} \ll NPV_{2024-2072}^{Deficits, SS} = 64\%-215\% \text{ of GDP.}$$

Both bounds lie well above the public sector’s net asset position of 45.9% of GDP.

Even in this scenario, the current net assets on hand, amounting to 45.9% of GDP, fall short of the lower-bound social security deficit of 64% of GDP—although the gap is far smaller than under the upper-bound valuation. This simple extended accounting exercise nonetheless suggests that the fiscal burden of social security would be substantially alleviated if pension benefits were indexed to Korea’s GDP. Importantly, this adjustment is not equivalent to reducing the benefit stream; rather, indexing benefits to GDP ensures that the benefit stream comoves with economic growth.

In short, there appears to be a silver lining. The fiscal concerns surrounding social security deficits caused by an aging population could be greatly alleviated if pension benefits were fully indexed to GDP growth. However, our simple exercise also reveals that the bleak fiscal outlook extends beyond the social security system. The central government cannot sustain its current spending plans even if social security is excluded. The Korean government must address its own ongoing and persistent fiscal deficits, which are currently masked by the temporary surplus in the social security system.

4.3 A VAR-Based Simulation of the Consolidated Balance Sheet

We quantitatively assess Korea’s long-term fiscal sustainability under the government’s carry-trade strategy from the perspective of the consolidated public-sector balance sheet, projecting it forward with a VAR-based Monte Carlo simulation. The exercise has two ingredients: we first forecast the return on every asset and liability category, and then feed these returns—together with NABO’s projected deficits—into a simulation that traces the year-by-year evolution of the consolidated balance sheet through 2072 under our baseline financing rule, in which deficits are financed by government bond issuance.

Forecasting the Returns We treat the two kinds of return differently. Market-determined returns—on domestic and foreign equities, bonds, and reserves—are forecast *stochastically* with the VAR, which generates a full distribution of future paths. Returns on policy-controlled instruments—central bank securities, deposits, bank reserves, and government loans—are instead projected *de-*

terministically, from NABO’s official base-rate forecast (a Taylor-rule projection) plus each instrument’s fixed historical spread. Table 8 classifies every return accordingly. The government bond yield, which governs the funding cost on public debt, is itself one of the market-determined series forecast stochastically by the VAR; we do not impose NABO’s deterministic, gradually declining yield path on the simulation.

Table 8: Stochastic versus Deterministic Return Projections

Balance-sheet return	Proxy / index
<i>Stochastic — forecast by the VAR (Monte Carlo, 7 series)</i>	
Government bond return	KIS Government Bond Index
Corporate bond return	KIS Corporate Bond Index
Domestic equity return	KOSPI Index
Foreign equity return	MSCI ACWI (ex Korea)
Foreign bond return	Global fixed-income index
Hedge fund return	Credit Suisse HF Index
Foreign reserve return	ICE BofA 5–7Y US Treasury
<i>Deterministic — NABO base-rate path + fixed historical spread</i>	
Deposits (net)	Liquidity Adjustment Deposit Rate
BOK loans	Standing Lending Facility
Bank reserves	Bank reserve rate
Central bank securities	91-day Monetary Stabilization Bond rate
Government loans	PFMF lending rate
Currency	Zero return

Note: The seven market-determined return series are forecast stochastically by the VAR through Monte Carlo simulation (Table 9); the policy-controlled returns are projected deterministically from NABO’s base-rate forecast plus each instrument’s fixed historical spread over 2000–2023. Each proxy is defined in Table 7.

Source: Authors’ construction.

The VAR is specified to capture the characteristics of Korea as a small open economy. It comprises eleven endogenous variables—seven market-determined return series and four macroeconomic indicators—listed with their proxies and sources in Table 9. All foreign-asset returns are expressed in Korean won terms to maintain measurement consistency within the system.

We estimate the VAR on quarterly data from 2003Q1 to 2024Q4 with a lag length of four, and then simulate 5,000 forward trajectories over 2025–2072 for the seven market-determined return series. Consistent with standard VAR dynamics, the projected returns mean-revert toward their long-run unconditional means while the surrounding uncertainty widens over the horizon; detailed fan charts are reported in Appendix E. Because the unrestricted (OLS) estimates have a largest eigenvalue close to unity in a sample of only 88 quarters, the implied long-run means are weakly identified and the projected paths are sensitive to estimation noise. As a robustness check, Appendix F re-estimates the system with a Bayesian VAR under a Minnesota prior, which shrinks the dynamics toward greater stability; the resulting return spread is modestly larger and the projected fiscal deterioration somewhat slower, but the qualitative conclusion of long-run fis-

Table 9: Endogenous Variables in the VAR Model

Variable	Proxy / Index	Source
<i>Market return series (7)</i>		
Foreign bond return	Global fixed-income index	Bloomberg
Foreign equity return	MSCI ACWI (ex Korea)	Bloomberg
Hedge fund return	Credit Suisse HF Index	Bloomberg
Government bond return	KIS Government Bond Index	Bloomberg
Corporate bond return	KIS Corporate Bond Index	Bloomberg
Domestic equity return	KOSPI Index	Bloomberg
Foreign reserve return	ICE BofA 5–7Y US Treasury	Bloomberg
<i>Macroeconomic variables (4)</i>		
World production growth	Global production index	Baumeister & Hamilton (2019)
GDP per worker growth	Real GDP per worker	Bank of Korea (ECOS)
CPI inflation	Consumer Price Index	Bank of Korea (ECOS)
KRW/USD exchange-rate change	Won-dollar spot rate	Bank of Korea (ECOS)

Note: The VAR is estimated on quarterly data from 2003Q1 to 2024Q4 with a lag length of 4. The global production index follows Baumeister and Hamilton (2019); its data are available at <https://sites.google.com/site/cjsbaumeister/datasets>. The construction of each return index is detailed in Table 7.

cal unsustainability is unchanged. Finally, because it is estimated on historical data, the VAR does not endogenously embed a sovereign-risk premium that rises with the debt level; such a premium would raise funding costs and, if anything, reinforce our conclusion of long-run fiscal unsustainability, so our baseline is conservative in this respect.

Balance Sheet Evolution and Deficit Financing We project the consolidated balance sheet forward year by year, financing the projected deficits by a simple rule. Deficits of the social-security system (the NPS) are met by drawing down the fund’s own asset holdings until those reserves are exhausted, while every other general-government deficit is financed by issuing government bonds; once the NPS fund is depleted, the remaining pension shortfalls also shift to bond financing. The deficit paths come from NABO’s projections, except that for the NPS we apply its post-reform revenue and benefit paths under the 2025 *National Pension Act reform*, rather than the frozen no-reform baseline of Table 2.⁹

Referring to Table 2, the operational surplus of the social security system in 2025 is projected at

⁹The 2025 reform raises the contribution rate from 9% to 13% and the income replacement rate from 40% to 43%, and expands childbirth and military-service credits; we apply NABO’s own post-reform NPS revenue and benefit paths. Relative to the frozen baseline of Table 2, the reform improves the 2072 fiscal aggregates as follows:

Fiscal aggregates in 2072: baseline versus 2025 reform		
2072, % of GDP	Baseline	With reform
Social-security balance	–5.2	–4.9
operational (excl. investment income)	–5.2	–4.9
Total deficit	11.6	10.8
excl. interest	8.8	8.0
excl. SS & interest	3.2	3.2

Source: NABO Long-Term Fiscal Projections 2025–2072.

The higher contributions also delay the projected depletion of the NPS fund; the asset-side and market-return inputs are unaffected by the reform.

0.4 percent of GDP, while the total operational balance of the general government is zero. Under our financing rule, NPS assets in the consolidated balance sheet—currently 46 percent of GDP—would rise by 0.4 percentage points. The government’s liabilities would increase by the remaining 0.4 percent deficit, reflecting the gap between the total balance and the social security surplus. In 2030, the social security system is projected to record an operational deficit of 0.4 percent of GDP, while the total deficit is 0.7 percent of GDP. In this case, the NPS fund size would decline by 0.4 percentage points, whereas government bond liabilities would increase by 1.1 percent of GDP, the sum of the social security deficit and the total deficit.

The simulation starts from the consolidated balance sheet at the end of 2024 and tracks annual asset and liability positions along each of the 5,000 simulated paths. On the asset side, we disaggregate the holdings into the National Pension Service (NPS), the Korea Investment Corporation (KIC), and the remaining general-government and Bank of Korea holdings, and grow each block at its own portfolio return—rather than a single blended rate—holding each portfolio’s composition at its 2024 GDP shares. The NPS’s and KIC’s globally diversified holdings earn market returns on their respective asset classes, whereas the general government’s strategic domestic-equity stakes (public-corporation and fund shares, 18.1 percent of GDP) earn only their realized dividend yield, since these positions are policy-held and are not liquidated for capital gains. On the liability side, government bonds expand to finance the annual deficits, while the remaining liability components grow at their own return rates. Each year, every stock is then rescaled by NABO’s projected nominal GDP growth, defined as the sum of real GDP-per-worker growth, employment growth, and CPI inflation; the first and third are taken from the VAR forecast, while employment growth follows NABO’s long-term projection.¹⁰

Simulation Results. Over the forecast horizon of 2025–2072, the simulation yields an average asset–liability spread of 1.9 percentage points. Figure 4 illustrates the forecasted trajectories of asset returns and the resulting spreads. Subplot (a) displays the median return paths for both assets and liabilities, with the red solid line indicating the median asset return and the blue solid line representing the median liability return. The shaded bands denote the 95% confidence interval (2.5th to 97.5th percentile) of the return distribution across 5,000 simulated paths.

Initially, asset returns exhibit considerable volatility, primarily driven by the large portfolio shares of risky instruments such as domestic and foreign equities. However, consistent with the mean-reverting nature of the VAR forecast, asset returns gradually converge toward their long-run average of approximately 4.1% over time. In contrast, liability returns display a more stable trajectory, reflecting the relatively lower volatility of policy-controlled instruments and government bonds. These returns steadily converge to around 2.2% in the long run. As a result, the

¹⁰According to NABO, the number of employed persons is projected to peak at 29.08 million in 2032 and decline thereafter due to falling population growth and labor force participation rates. By 2072, employment is expected to fall to 18.28 million. This implies an annual employment growth rate that rises until 2032 (peaking at 0.21%) and subsequently declines to -1.46% by 2072.

Figure 4: Projected Returns and Spreads on the Consolidated Balance Sheet



Note: Shaded areas indicate the 95% confidence intervals; solid lines represent median values.

asset–liability spread in subplot (b) stabilizes near 1.9 percentage points in the later years of the simulation period.

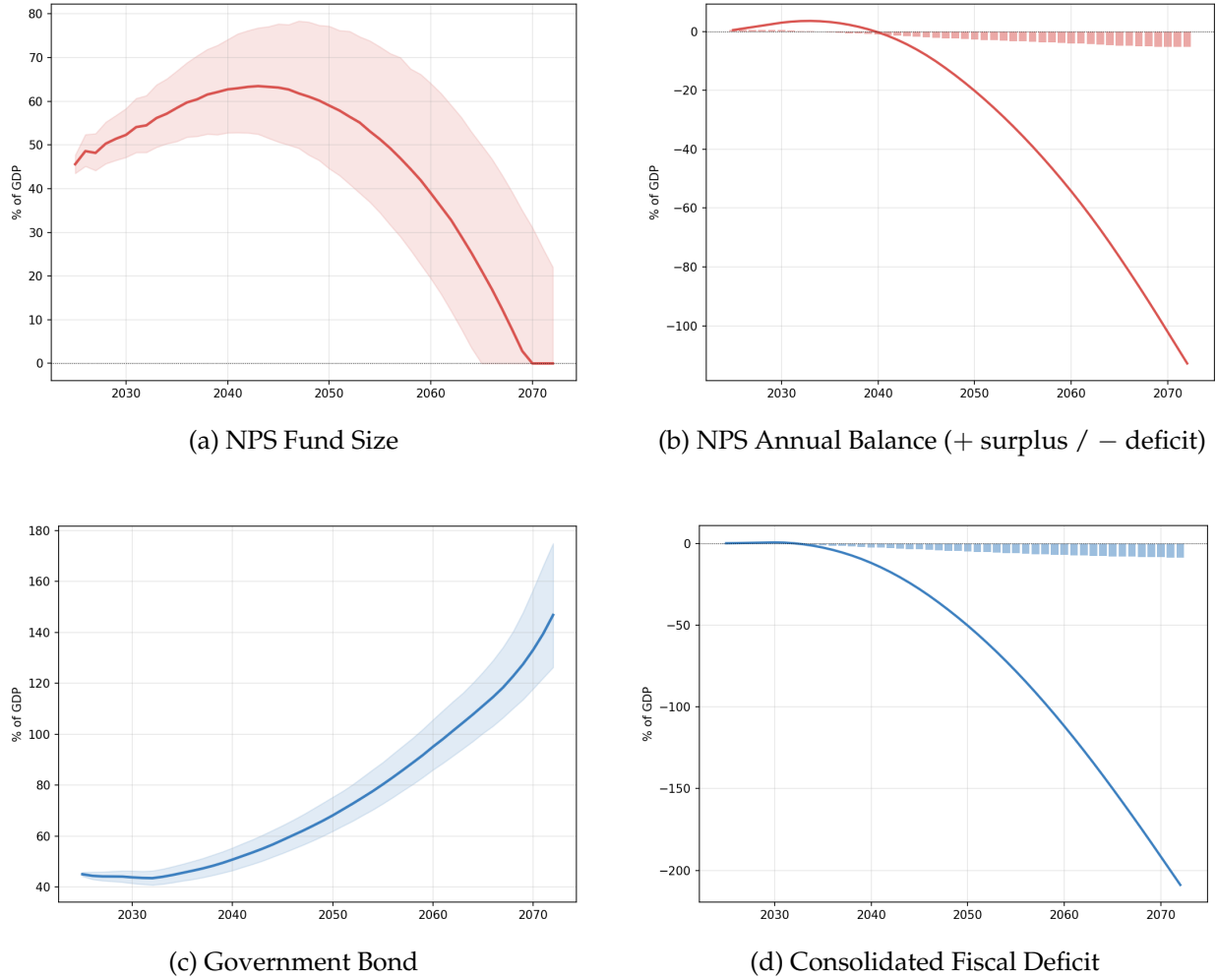
Persistent fiscal deficits significantly influence the evolution of both the asset and liability sides of the consolidated government balance sheet. In particular, annual deficits gradually deplete the National Pension Service (NPS) Fund and drive a sharp increase in government bond liabilities. Figure 5 presents the projected time paths for the size of the NPS Fund and its deficits (subplots a and b), as well as for government bonds and overall fiscal deficits (subplots c and d).

On the asset side, the NPS Fund initially expands, peaking at roughly 63% of GDP in the early 2040s, reflecting positive early cash flows under the higher contribution rate of the 2025 pension reform. However, as shown in subplot (b), persistent pension deficits subsequently erode the fund. According to the median simulation path, the NPS Fund becomes fully depleted by 2070. This timing is consistent with the National Assembly Budget Office’s own sensitivity analysis, which dates depletion to 2071 under a 5.5% fund return—close to the consolidated public sector’s market (carry-trade) return in our simulations—rather than the 2065 headline figure, which assumes a more conservative 4.5% return. Once this depletion occurs, all subsequent pension-related shortfalls are covered by general government borrowing.

On other side, government bond liabilities increase rapidly. Subplot (c) shows that the government bond-to-GDP ratio approaches 150% by 2072 under the baseline simulation. This trajectory is largely driven by two compounding forces: the accumulation of general fiscal deficits and the additional burden from covering pension deficits after the NPS Fund is exhausted. The simulation clearly highlights a sharp inflection in the debt path beginning around 2070, corresponding to the transition point where pension liabilities are no longer asset-financed but instead fully debt-financed through government bond issuance.

Despite the positive return spread generated by the government’s carry trade strategy, persis-

Figure 5: Projections of Key Fiscal and Pension Indicators

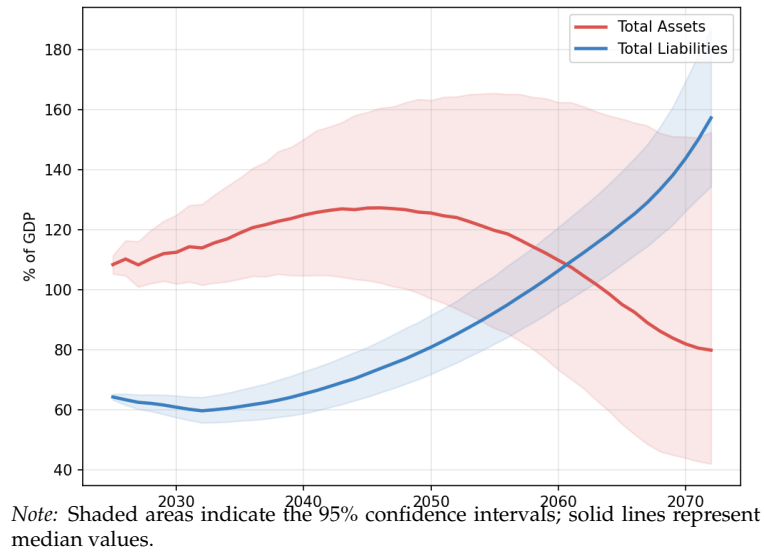


Note: Shaded areas indicate the 95% confidence intervals; solid lines represent median values.

tent structural deficits ultimately lead to a deterioration in the government's net asset position. As shown in Figure 6, total assets (red line) and total liabilities (blue line) follow markedly different trajectories over the forecast horizon. On the asset side, growth is constrained by the depletion of the NPS Fund and limited accumulation beyond its initial composition. In contrast, liabilities increase rapidly over time, fueled by the accumulation of fiscal deficits and the transition of pension financing from assets to debt.

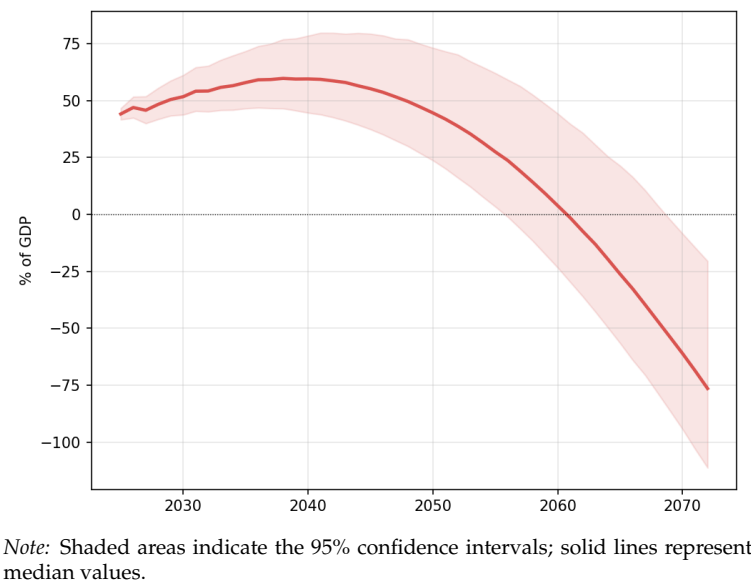
These diverging trends culminate in a net asset turning point around 2061, when median projected liabilities surpass total assets. Figure 7 illustrates the projected path of the net asset position as a share of GDP. Although the government benefits from a stable return spread of approximately 1.9 percentage points—driven by higher returns on risky assets relative to low-cost liabilities—the compounding effect of continuous deficits overwhelms the gains. As a result, the net asset position turns negative around 2061 and continues to deteriorate sharply. By 2072, the median net

Figure 6: Projected Asset and Liability Levels



asset position reaches approximately -76% of GDP. Consolidating the public sector's risky assets against its liabilities thus does not avert the underlying fiscal deterioration: even crediting the public funds with their full market returns, the carry trade only delays and moderates—rather than prevents—a deeply indebted terminal state.

Figure 7: Projected Net Asset Position



5 Financial Repression as an Alternative Policy

Faced with the deep fiscal shortfall documented above, a government has a tempting alternative to painful reform: *financial repression*, the practice of borrowing from captive domestic investors at below-market rates (Reinhart et al., 2011). Such policies are neither new nor confined to developing economies. Advanced economies leaned on them to work down their war debts through the 1970s—negative real interest rates liquidated government debt at three to four percent of GDP a year in the postwar United States and United Kingdom (Reinhart and Sbrancia, 2015; Sargent et al., 2019; Reinhart and Rogoff, 2009; Acalin and Ball, 2022)—and the term itself was coined to describe Korea’s own interest-rate ceilings and directed credit in the 1960s and 1970s (McKinnon, 1973). By channeling funds to the public sector on terms a free market would not offer, repression lowers the government’s borrowing cost and, with it, the apparent burden of its debt.

Korea is unusually well-placed to pursue this route, because its public sector already owns the captive buyer that repression requires. The classic instrument of repression is a large domestic institution obliged to hold government debt at suppressed yields (Reinhart and Sbrancia, 2015), and Korea’s National Pension Service—at about 46% of GDP the largest pool of domestic savings and a major holder of Korean Treasury bonds—is precisely such an institution, complemented by the reserves of the Bank of Korea and the assets entrusted to the KIC. The very consolidation that makes Korea’s carry trade possible thus also equips it to repress, which makes repression a live policy alternative rather than a hypothetical one and warrants taking its promise seriously.

5.1 Japan’s Experience

Japan shows what this strategy looks like in practice. Facing the same aging-driven fiscal pressures that Korea now confronts, Japan under Abenomics turned its public pension system into a de facto sovereign wealth fund financed by government borrowing, and paired this with aggressive monetary support—quantitative easing and, from 2016, yield-curve control—that let the government fund its debt at persistently low cost. Whatever their stated aim of reflation, these measures amount to a modern form of financial repression. Japan reached this point from a very different starting position than Korea, with central-government debt already above 183% of GDP by 2012;¹¹ Korea, by contrast, has built its risky asset position before accumulating comparable debt.

This repression-financed carry trade has visibly improved Japan’s fiscal position. Studying Japan’s consolidated balance sheet, Chien et al. (2025a) find that the public sector earns a substantial spread over its funding cost—more than six percent of GDP a year over the past decade—so that, despite continued large deficits, Japan’s consolidated net liabilities have actually declined. On its own terms, in other words, the strategy appears to have worked.

¹¹FRED, [Central Government Debt for Japan](#).

That apparent success, however, rests on an implicit tax rather than on any creation of resources. As [Chien et al. \(2025a\)](#) emphasize, Japan’s spread depends on persistently below-market interest rates, which levy a hidden charge on the captive depositors and bondholders obliged to hold low-yielding claims—a transfer from financially unsophisticated households to more sophisticated ones, and ultimately to the government. The strategy also leans heavily on the central bank, straining the separation between monetary and fiscal policy. Japan’s experience thus poses the question sharply for Korea: how much can repression actually deliver, and at whose expense?

5.2 Simulating Financial Repression

The literature on financial repression points to a specific, relatively mild form as the one advanced economies actually adopt at debt levels like Korea’s. [Jeanne \(2025\)](#) distinguishes two forms of increasing severity. In the first, *balance-sheet* repression, banks and the central bank are made to absorb the government debt that private investors will no longer hold, but they do so at market rates and the government extracts no revenue from them; in the second, *quasi-fiscal* repression, they are made to hold that debt at *below-market* rates, so that the government collects a flow of revenue $\theta = (r - r_\ell)\ell$ from its captive creditors. Empirically, advanced economies enter the balance-sheet stage once gross debt crosses roughly 100 to 120 percent of GDP, but no country—not even Japan, at over 200 percent—has yet resorted to the quasi-fiscal stage, which Jeanne characterizes as a last resort. The balance-sheet form triggered near 100 percent of GDP is therefore the empirically grounded baseline, and we take it as our starting point.

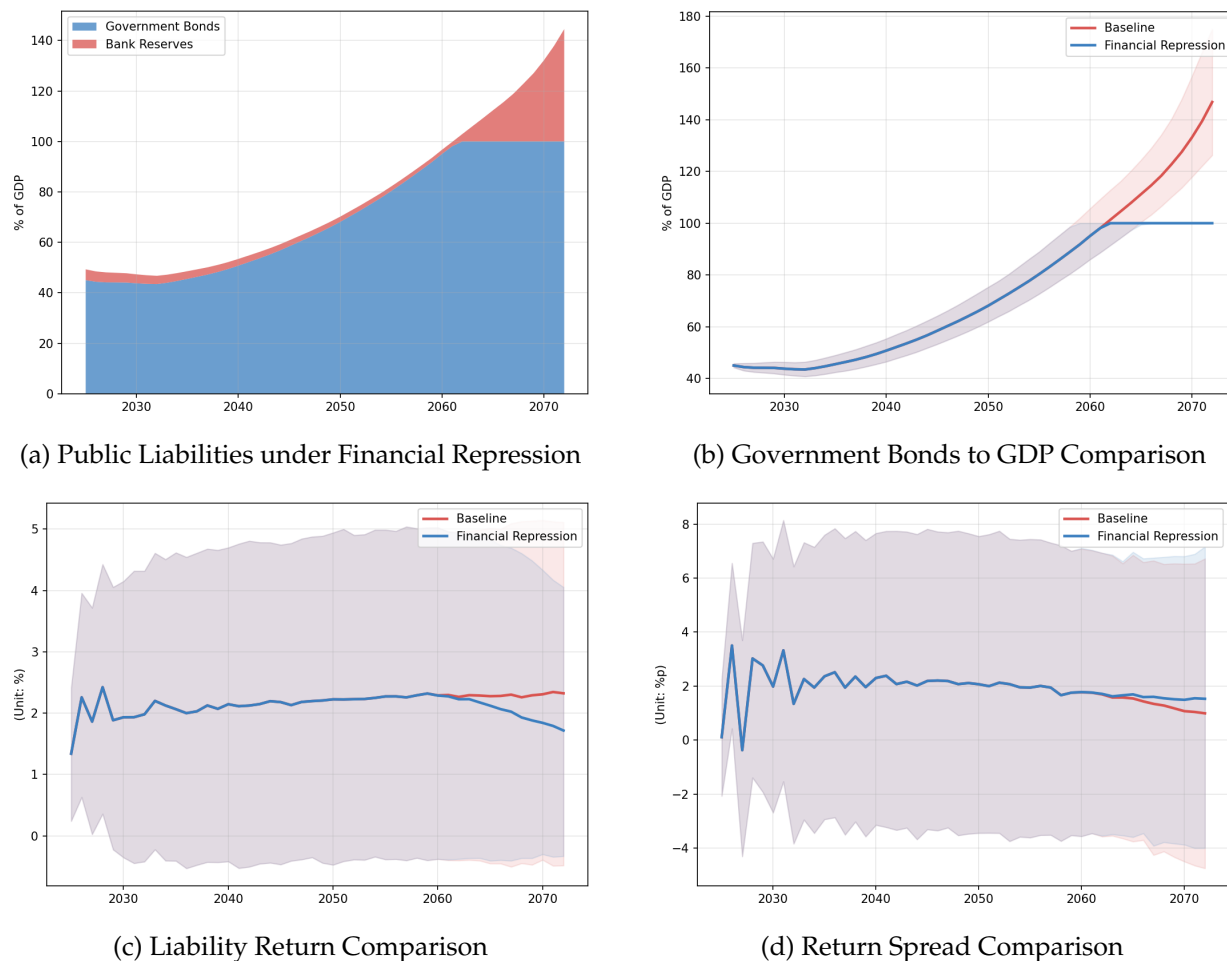
We implement this baseline form directly in the simulation. Once the government’s marketable bonds reach 100 percent of GDP—Jeanne’s lower threshold, which the median projection crosses around 2062—we cap them there and finance every subsequent deficit by issuing central-bank reserves rather than bonds. Because the interest paid on reserves at the Bank of Korea is effectively zero, substituting reserves for bonds lowers the average cost of the public sector’s liabilities; this is the channel through which balance-sheet repression works, and the whole of the relief it delivers comes from this reduction in funding cost.

Whatever its form, our exercise gives repression its most favorable possible reading. We hold the macroeconomic environment fixed—growth, asset returns, and inflation follow the same simulated paths as in the baseline—and change only the rate the government pays on its debt. In general equilibrium, repression is not costless: by forcing captive institutions to hold low-yielding public debt it crowds out their lending and distorts the allocation of capital, lowering growth ([Reis, 2021](#); [Jeanne, 2025](#)), which would feed back into a worse debt trajectory. Our estimates therefore overstate what repression can achieve, so that if even this best case fails to restore sustainability, the conclusion is only reinforced once these costs are admitted.

Figure 8 traces how the baseline program works. Panels (a) and (b) show the change in composition: once government bonds reach 100 percent of GDP their stock is held constant while bank reserves accumulate in their place (panel a), so that the bond ratio is capped at 100 percent un-

der repression even as the baseline ratio climbs toward 147 percent by 2072 (panel b). Panels (c) and (d) show the fiscal consequence: because zero-cost reserves progressively replace low-yield bonds, the average return the public sector pays on its liabilities falls modestly below the baseline (panel c), and the asset–liability spread widens by the same modest amount (panel d). The effect is small precisely because the consolidated public sector already borrows cheaply, so replacing low-yield bonds with zero-cost reserves saves comparatively little.

Figure 8: Mechanics of Balance-Sheet Repression



The relief this buys is slight: the program delays the fiscal deterioration but cannot prevent it. The average liability cost falls by only about a quarter of a percentage point, total liabilities in 2072 edge down from roughly 157 to 153 percent of GDP, and the terminal net asset position improves only from -76 to -73 percent. Net worth still turns negative around 2061 and continues to fall. This is exactly Jeanne’s balance-sheet stage, in which debt is absorbed at market rates and no revenue is extracted; at Korea’s projected debt level, it is very nearly powerless.

Is this feeble result merely a matter of degree—is the program simply too mild? To answer this, we turn to the aggressive, quasi-fiscal form, in which the government pushes the real return on its debt *below* the market rate by a wedge ω , collecting the revenue $\theta = (r - r_\ell)\ell$ from its captive creditors. We vary this program along both dimensions that could make it more potent: the size of the wedge and the date at which it is adopted.

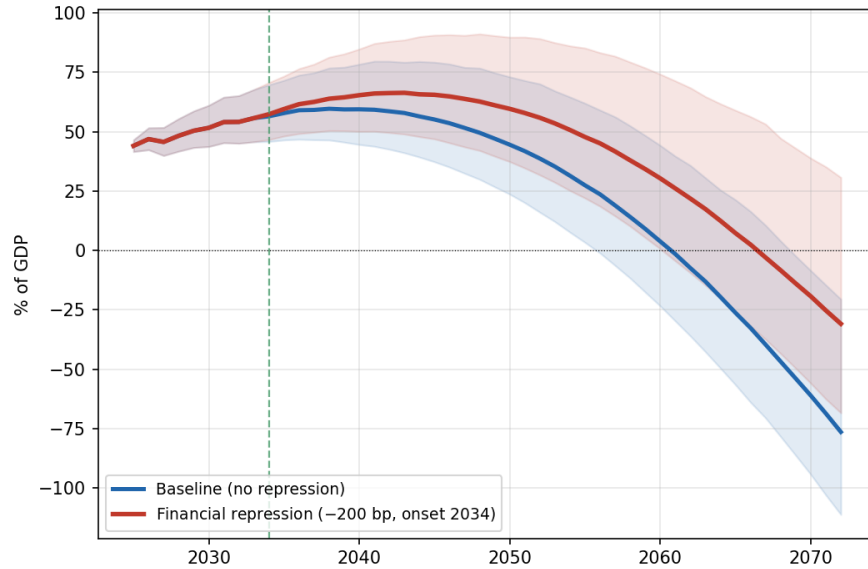
Such a program is not hypothetical; it has ample historical precedent. In the three decades after the Second World War, the advanced economies liquidated their war debts in precisely this way, holding real interest rates below zero long enough to erase three to four percent of GDP a year in the United States and the United Kingdom (Reinhart and Sbrancia, 2015). Japan does much the same today: Chien et al. (2025a) estimate that its consolidated public sector pays a below-market wedge of about -2.2 percent on its debt. The instrument in every case is a captive domestic holder obliged to accept low yields—exactly the role Korea’s National Pension Service is positioned to play.

That same precedent, however, bounds how far repression can credibly be pushed. Reis (2022, 2025) show that the revenue repression can raise is limited and, in Reis’s words, will not accrue “systematically and persistently”; the durable ceiling is on the order of the three-to-four-percent-of-GDP liquidation that Reinhart and Sbrancia document. We therefore set the central wedge at -200 basis points, close to Japan’s realized figure, and report -300 and -400 basis points as progressively stronger variants—the last already beyond what the literature regards as sustainable. Each wedge is applied under two timings: at the 100-percent debt trigger around 2062, and, as an upper bound on how early the policy could plausibly begin, from the onset of Korea’s fiscal deadlock in 2034, by which its social-security *operating* balance—contributions net of benefits, before investment income—has moved into deficit.

Even at these stronger settings, repression still falls short. Table 10 reports the terminal net positions; Figure 9 traces the path under the central program and Figure 10 under the stronger variants, with the full set of results—including the return spread and the paths of liabilities and debt—collected in Appendix B. Applied at the debt trigger, the central -200 -basis-point wedge lifts the 2072 net asset position from -76 to -55 percent of GDP, and the stronger -300 and -400 wedges to -47 and -38 percent; adopting repression early, in 2034, roughly halves the shortfall again, to -31 , -16 , and -5 percent respectively. Within the range the literature regards as durable—wedges up to about -350 basis points, liquidating no more than three to four percent of GDP a year—the terminal net position remains deeply negative in every case. Even the -400 -basis-point wedge adopted in 2034, which lies beyond that durable range, only reaches -5 percent of GDP: repression at its most aggressive brings net worth to the edge of zero but not across it.

The lesson is the same across the whole range: financial repression can postpone Korea’s fiscal reckoning but cannot resolve it. The failure is not one of insufficient intensity, since within any sustainable calibration net worth still ends deeply negative; it is that repression, as the next subsection shows, creates no new resources and merely relocates the fiscal burden. It is, in the end,

Figure 9: Projected Net Asset Position: Baseline vs. Financial Repression



Note: Median projected net asset position (% of GDP): the no-repression baseline versus the central quasi-fiscal program—a -200 -basis-point wedge adopted at the 2034 fiscal-deadlock onset (vertical line). The shaded bands are the 95% confidence intervals for the baseline (blue) and repression (red) paths. Repression lifts the terminal position from -76 to -31 percent of GDP but leaves it deeply negative.

Table 10: Terminal Net Asset Position under Financial Repression (median, % of GDP in 2072)

	Repression wedge ω		
	-200 bp	-300 bp	-400 bp
<i>Reference:</i> baseline (no repression) -76 ; balance-sheet form -73			
Quasi-fiscal, trigger 2062 (100% debt)	-55	-47	-38
Quasi-fiscal, onset 2034 (fiscal deadlock)	-31	-16	-5

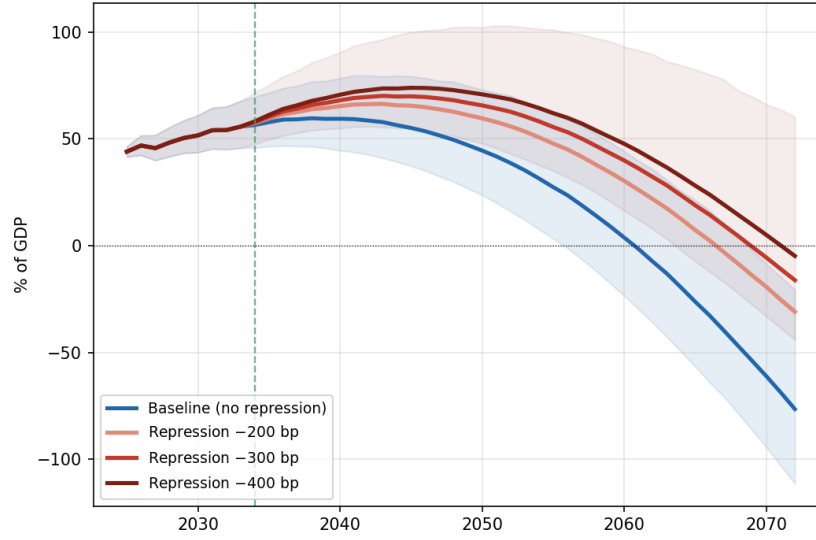
Note: Median terminal net asset position (% of GDP) across 5,000 simulations. The balance-sheet form caps bonds at 100% of GDP and finances the rest with zero-cost reserves; the quasi-fiscal wedges push the debt's real return below market by ω . The -400 -bp wedge lies beyond the range the literature regards as durable. Detailed results (return spread, liabilities, and debt) are reported in Appendix B.

no substitute for fundamental fiscal and pension reform.

5.3 The Incidence of Repression

Repression's limited relief is no accident, because it creates no resources; it only relocates a burden, and it does so with leverage. Return to the risk-premium identity of equation (1), which splits the excess return on the public sector's assets between the return paid to its bondholders and the return borne by the surplus claim. Lowering the debt return by the wedge ω , and holding the

Figure 10: Net Asset Position under Higher-Intensity Repression



Note: Median net asset paths under repression wedges of -200 , -300 , and -400 basis points, all adopted at the 2034 fiscal-deadlock onset (vertical line), against the no-repression baseline (blue). Shaded bands are the 95% confidence intervals for the baseline and the strongest (-400 bp) path. Even the -400 -basis-point wedge—beyond the range the literature regards as durable—leaves the terminal net position at about -5 percent of GDP.

asset return fixed, the excess return loaded onto the surplus claim rises by exactly

$$\Delta \mathbb{E}_t [R_{t+1}^S - R_t^f] = \frac{D_t}{N_t} \omega, \quad (3)$$

where $N_t = A_t - D_t$ is the public sector's net worth. Appendix A derives this and verifies that the identity is preserved. The wedge that captive creditors—in Korea, above all the National Pension Service and the domestic banks—forgo does not disappear; the identity reassigns it, levered by the debt-to-net-worth ratio D_t/N_t , to whoever holds the residual claim on the public balance sheet. That ratio is about 1.5 today and grows without bound as net worth falls toward zero, so repression concentrates the largest claim on the public precisely when—as in our projection after the early 2060s—the net-worth cushion that was meant to absorb it has all but vanished.

Because this reassignment follows from an accounting identity alone, it is, if anything, an understatement of repression's true cost. Our simulation already grants repression its best case, holding growth and asset returns fixed; but repression is not costless in general equilibrium. By forcing captive institutions to hold low-yielding public debt, it crowds out their lending to the real economy and distorts the allocation of capital, lowering growth (Reis, 2021; Jeanne, 2025)—and slower growth feeds directly back into a worse debt trajectory. Allowing for these effects would only deepen the shortfall that our partial-equilibrium exercise already finds. Financial repression, in short, can postpone Korea's fiscal reckoning but not avert it, and it works only by taxing the very households—savers, depositors, and pension contributors—that the public sector exists to

serve. It redistributes the cost of inaction; it is no substitute for fundamental fiscal and pension reform.

6 Conclusion

Korea is undergoing a rapid demographic transition into an aging society. In the foreseeable future, without significant social security reform and improved fiscal discipline, the government will be compelled to run large deficits in order to honor its commitments to the elderly population.

In the absence of extensive pension reform, the public sector's consolidated balance sheet has increasingly taken the form of a large carry trade—whether by deliberate design or, as we have argued, as the byproduct of the separate investment mandates of the NPS and the KIC—generating additional returns by assuming greater financial risk. Our theoretical analysis shows that for the balance sheet to earn an excess return, the additional risk must ultimately be borne by the government's bondholders or its taxpayers. In practice, however, shifting that risk onto taxpayers may become politically unfeasible once the government's fiscal capacity is exhausted.

We then conduct a numerical simulation to examine the likelihood of deteriorating fiscal space, as measured by the debt-to-GDP ratio. The results indicate a significant probability that Korea's fiscal situation could worsen substantially. Crediting the public sector with the full market returns on its risky assets, the carry trade only delays and moderates—rather than prevents—a deeply negative terminal net asset position. We therefore conclude that such financial strategies are no substitute for fundamental fiscal and pension reform.

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A Risk-Premium Equivalence and the Incidence of Repression

The derivation of equation (1) is a straightforward extension of [Jiang et al. \(2019\)](#). Denote Q_t as the nominal payouts (dividends, coupons, etc.) of government financial assets, and denote M as the pricing kernel. Let $P_t^S \equiv \mathbf{E}_t \left[\sum_{j=0}^{\infty} M_{t,t+j} S_{t+j} \right]$ represent the price of the government surplus claim. The government also holds assets, $A_t = \mathbf{E}_t \left[\sum_{j=0}^{\infty} M_{t,t+j} Q_{t+j} \right]$. The present value of the government budget constraint at period $t + 1$ is:

$$D_{t+1} = P_{t+1}^S + A_{t+1}.$$

Dividing the budget constraint equation by $D_t - S_t - Q_t$ gives:

$$\frac{D_{t+1}}{D_t - S_t - Q_t} = \frac{P_{t+1}^S}{D_t - S_t - Q_t} + \frac{A_{t+1}}{D_t - S_t - Q_t}.$$

We then have the following return equation:

$$\frac{D_{t+1}}{D_t - S_t - Q_t} = \frac{P_t^S - S_t}{D_t - S_t - Q_t} \frac{P_{t+1}^S}{P_t^S - S_t} + \frac{A_t - Q_t}{D_t - S_t - Q_t} \frac{A_{t+1}}{A_t - Q_t},$$

which can be rewritten as:

$$\mathbf{E}_t \left[R_{t+1}^D \right] = \frac{P_t^S - S_t}{D_t - S_t - Q_t} \mathbf{E}_t \left[R_{t+1}^S \right] + \frac{A_t - Q_t}{D_t - S_t - Q_t} \mathbf{E}_t \left[R_{t+1}^A \right],$$

where the returns are defined as:

$$R_{t+1}^D \equiv \frac{D_{t+1}}{D_t - S_t - Q_t}, \quad R_{t+1}^S \equiv \frac{P_{t+1}^S}{P_t^S - S_t}, \quad \text{and} \quad R_{t+1}^A \equiv \frac{A_{t+1}}{A_t - Q_t}.$$

Using $D_t = P_t^S + A_t$, we can rewrite the equation above as:

$$\begin{aligned} \mathbf{E}_t \left[R_{t+1}^D - R_t^f \right] &= \frac{P_t^S - S_t}{D_t - S_t - Q_t} \mathbf{E}_t \left[R_{t+1}^S - R_t^f \right] + \frac{A_t - Q_t}{D_t - S_t - Q_t} \mathbf{E}_t \left[R_{t+1}^A - R_t^f \right] \\ &= \left(1 - \frac{A_t - Q_t}{D_t - S_t - Q_t} \right) \mathbf{E}_t \left[R_{t+1}^S - R_t^f \right] + \frac{A_t - Q_t}{D_t - S_t - Q_t} \mathbf{E}_t \left[R_{t+1}^A - R_t^f \right] \\ &\approx \left(1 - \frac{A_t}{D_t} \right) \mathbf{E}_t \left[R_{t+1}^S - R_t^f \right] + \frac{A_t}{D_t} \mathbf{E}_t \left[R_{t+1}^A - R_t^f \right], \end{aligned}$$

which is the equation (1).

The incidence of repression. Building on the identity just derived, we now obtain equation (3). Write the expected excess returns as $\pi^A \equiv \mathbf{E}_t[R_{t+1}^A - R_t^f]$, and π^D, π^S analogously, and let $\lambda \equiv D_t/A_t$. The risk-premium identity (1) is $\pi^A = \lambda\pi^D + (1 - \lambda)\pi^S$; solving for the surplus-claim

premium gives

$$\pi^S = \frac{\pi^A - \lambda\pi^D}{1 - \lambda}.$$

Financial repression lowers the return the government pays on its debt by the wedge ω , so $\pi^D \mapsto \pi^D - \omega$, while the asset return π^A and the leverage λ are held fixed. Substituting and separating terms,

$$\pi_{\text{rep}}^S = \frac{\pi^A - \lambda(\pi^D - \omega)}{1 - \lambda} = \frac{\pi^A - \lambda\pi^D}{1 - \lambda} + \frac{\lambda}{1 - \lambda} \omega = \pi^S + \frac{\lambda}{1 - \lambda} \omega.$$

Since $\frac{\lambda}{1 - \lambda} = \frac{D_t/A_t}{(A_t - D_t)/A_t} = \frac{D_t}{A_t - D_t} = \frac{D_t}{N_t}$ with $N_t \equiv A_t - D_t$, the change in the surplus-claim premium is $\Delta\pi^S = \frac{D_t}{N_t} \omega$, which is equation (3).

The identity is preserved. Substituting $\pi^D - \omega$ and $\pi^S + \frac{\lambda}{1 - \lambda} \omega$ into the right-hand side of (1) yields $\lambda(\pi^D - \omega) + (1 - \lambda)(\pi^S + \frac{\lambda}{1 - \lambda} \omega) = \lambda\pi^D + (1 - \lambda)\pi^S - \lambda\omega + \lambda\omega = \pi^A$: the excess return of $\lambda\omega$ per unit of assets that bondholders forgo is exactly the amount reassigned to the surplus claim, and the asset risk premium π^A is unchanged. Because ω shifts expected returns deterministically, the result is a levered transfer of excess return from captive creditors to the residual claimant rather than a change in the underlying quantity of risk.

B Financial Repression: Detailed Scenario Results

Table 11 reports the full set of median outcomes for the repression scenarios discussed in Section 5: the terminal net asset position, the average asset–liability return spread over the horizon, the 2072 levels of total liabilities and of government bonds, the year the net asset position first turns negative, and the year the National Pension Service fund is depleted. The spread rises monotonically with the wedge—from 1.9 percentage points in the baseline to 4.5 under the strongest program—confirming that repression operates entirely through the liability side; yet in no sustainable case does the terminal net position climb back out of deficit.

C Separate Balance Sheets and Returns

This appendix reports the balance sheets of the two main components of the consolidated public sector on their own, before consolidation. Table 12 presents the general government—which in the national accounts already subsumes the social security fund managed by the NPS—and Table 13 presents the Bank of Korea. Consolidating these two sectors together with the foreign assets managed through the KIC, and netting out the claims that public entities hold on one another, yields the consolidated balance sheet reported in Table 3 of the main text.

The two sectors play very different roles. The general government is where the risk-taking is concentrated: its assets rose from 53.8% of GDP in 2008 to 86.3% in 2024, an increase driven almost entirely by foreign assets (from 2.0% to 35.9% of GDP, of which foreign equities alone climbed

Table 11: Financial Repression Scenarios: Detailed Median Results

Scenario	net@2072 (% GDP)	spread (%p)	liab. (% GDP)	gov. bonds (% GDP)	net<0 (year)	NPS depl.
Baseline (no repression)	−76	1.89	157	147	2061	2070
Balance-sheet (100% cap, reserves)	−73	1.94	153	100	2061	2070
<i>Quasi-fiscal, trigger at 100% debt ($\approx$ 2062):</i>						
−200 bp	−55	2.26	136	100	2061	2070
−300 bp	−47	2.44	127	100	2061	2070
−400 bp	−38	2.62	118	100	2061	2070
<i>Quasi-fiscal, onset at fiscal deadlock (2034):</i>						
−200 bp	−31	3.23	111	100	2067	2070
−300 bp	−16	3.87	96	85	2069	2070
−400 bp	−5	4.48	84	74	2072	2070

Note: Median across 5,000 simulations, 2025–2072. “spread” is the average asset–liability return spread over the horizon; “net<0” is the year the net asset position first turns negative; “NPS depl.” is the year the National Pension Service fund is exhausted. The −400-bp wedge lies beyond the durable range documented by [Reinhart and Sbrancia \(2015\)](#) and [Reis \(2025\)](#).

from 1.1% to 22.3%), whereas its liabilities are almost all safe government bonds (38.2% of GDP in 2024). Its net asset position accordingly improved from 28.8% to 42.7% of GDP. The Bank of Korea, in contrast, holds a stable, liquidity-oriented portfolio—dominated by foreign-exchange reserves (16.9% of GDP in 2024)—funded by monetary liabilities such as currency, bank reserves, and central-bank securities, so that its balance sheet is roughly matched and its net position remains small and steady at about 3.3% of GDP.

Figure 11 summarizes the return performance of the consolidated balance sheet over 2008–2024. Panel (a) plots the estimated rates of return—the return earned on the asset side, the cost paid on the liability side, and the spread between them—while panel (b) rescales the same quantities by the size of each side to express the net return as a percent of GDP. Two features stand out. First, during the first half of the sample the spread was negligible: asset and liability returns moved closely together, so the public sector earned little from its balance sheet. Second, the spread widened markedly after 2015, as the portfolio shifted toward riskier foreign assets and as those assets—domestic and global equities in particular—delivered unusually strong market returns. The consequence, visible in panel (b), is a positive and rising net return in percent of GDP over the latter half of the sample: the carry documented in the main text.

D Validation of Return Proxies for NPS and KIC

This appendix validates the return series employed in our balance-sheet simulation by benchmarking them against the actual performance reported by Korea’s National Pension Service (NPS) and the Korea Investment Corporation (KIC).

Table 12: General Government Balance Sheet

% of GDP, Year End	2008	2015	2024	08 - 24 Diff
Assets				
Gold, SDR, Currency and Deposits	9.0%	8.4%	5.1%	-3.9%
Government Loans	12.2%	9.9%	10.0%	-2.2%
Domestic Debt Securities	6.7%	7.5%	7.3%	0.6%
Domestic Equities	13.3%	16.4%	15.8%	2.5%
Investment Fund Shares	4.3%	5.1%	6.5%	2.2%
Foreign Assets	2.0%	11.2%	35.9%	33.9%
Foreign Debt Securities	0.2%	0.2%	1.5%	1.3%
Foreign Equities	1.1%	6.8%	22.3%	21.2%
Other Foreign Claims	0.7%	3.1%	6.8%	6.1%
Foreign Direct Investment	0.0%	1.0%	5.3%	5.3%
Others	6.3%	6.0%	5.7%	-0.6%
Sum	53.8%	64.3%	86.3%	32.5%
Liabilities				
Government Bonds	19.9%	30.3%	38.2%	18.3%
Others	5.1%	5.9%	5.4%	0.3%
Sum	25.0%	36.2%	43.6%	18.6%
Net Assets	28.8%	28.1%	42.7%	13.9%

Note: Unit: % of GDP.

Source: Korean Flow of Funds and National Accounts of Korea.

Table 13: Bank of Korea Balance Sheet

% of GDP, Year End	2008	2015	2024	08 - 24 Diff
Assets				
Government Bonds	1.0%	1.0%	1.0%	0.0%
Domestic Equities	0.2%	0.1%	0.1%	-0.1%
FX Reserves	20.6%	21.6%	16.9%	-3.7%
Others	3.1%	1.7%	2.7%	-0.4%
Sum	24.9%	24.4%	20.7%	-4.2%
Liabilities				
Gold, SDR, Currency and Deposits	5.5%	9.8%	7.6%	2.1%
Bank Reserves	4.5%	3.7%	4.4%	-0.1%
Central Bank Debt Securities	10.8%	10.5%	4.5%	-6.3%
Others	0.7%	0.2%	0.9%	0.2%
Sum	21.6%	24.2%	17.4%	-4.2%
Net Assets	3.3%	0.2%	3.3%	0.0%

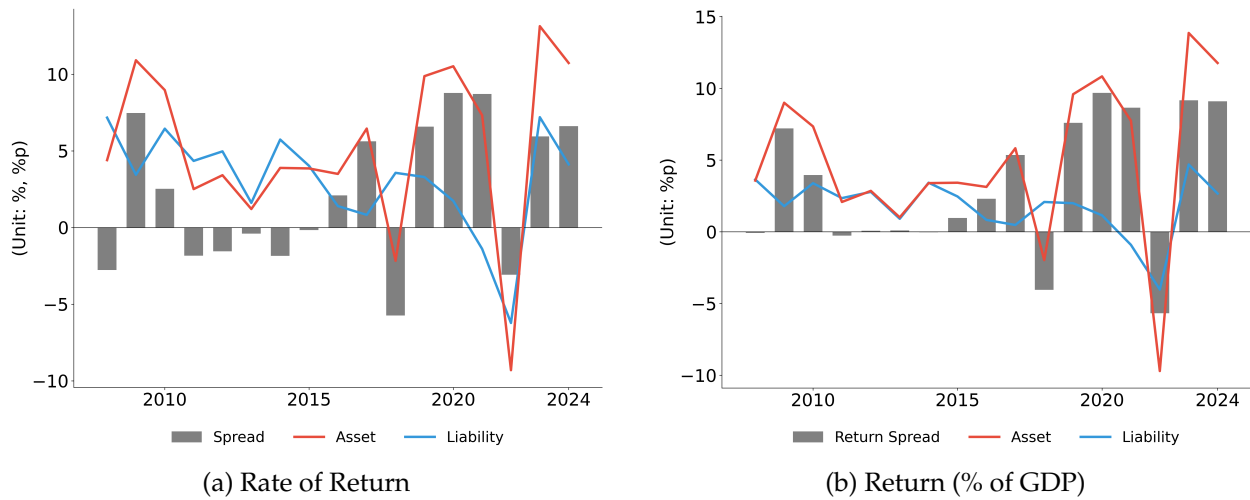
Note: Unit: % of GDP.

Source: Korean Flow of Funds and National Accounts of Korea.

Comparison with NPS Returns

Figure 12 juxtaposes the annual returns of our proxy indices with the official NPS figures across four asset buckets. For domestic equity (panel a), our KOSPI total-return series closely tracks

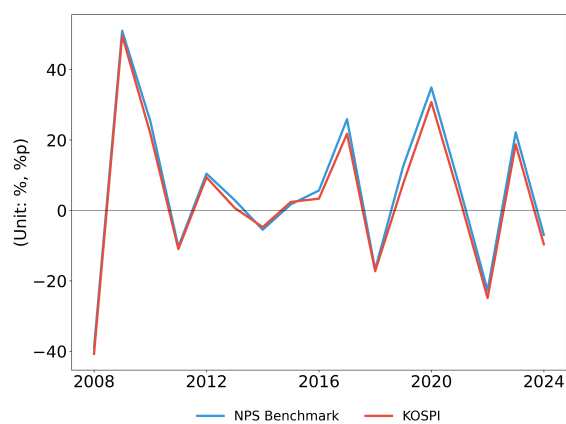
Figure 11: Estimated Return of Consolidated Balance Sheet



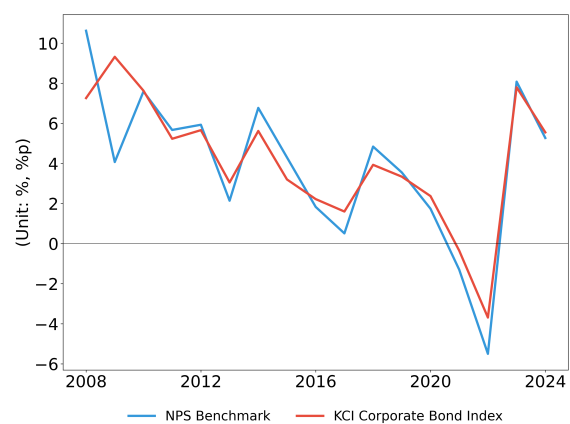
the NPS domestic-equity return in both direction and magnitude, and for domestic fixed income (panel b) the KIS Corporate Bond Index reproduces both the level and the cyclical pattern of NPS performance. On the foreign side, the MSCI ACWI index in unhedged won terms moves almost one-for-one with NPS global-equity returns (panel c), while for global fixed income (panel d) we reconstruct the NPS benchmark—the Bloomberg Aggregate Treasury Total Return Index (50%, hedged USD), the Bloomberg Global Aggregate–Government Index (20%), and the Bloomberg Global Aggregate–Corporate Index (30%)—and obtain a series virtually indistinguishable from the NPS realized returns. Across all four buckets the root-mean-square error between our proxies and the NPS outcomes stays small, generally below one percentage point per year, confirming that our indices faithfully capture the risk–return profile of the pension fund.

Comparison with KIC Returns

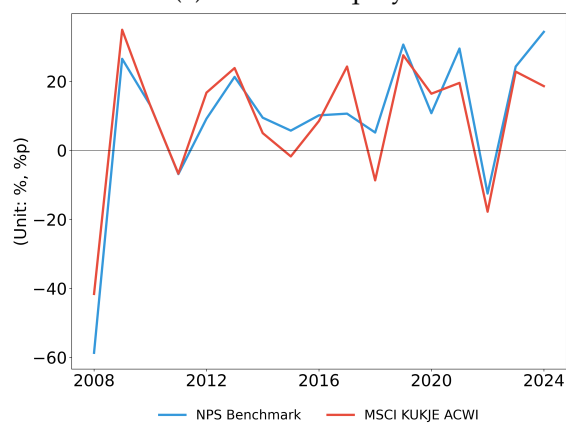
Figure 13 repeats the exercise for the KIC. For global equity (panel a), the MSCI ACWI index aligns closely with the KIC’s reported equity performance; minor level differences reflect the KIC’s active-management overlays, but the trends are well matched. For global fixed income (panel b) we employ the same customized global fixed-income benchmark used in the NPS comparison, and the resulting series again mirrors the KIC’s reported returns, indicating that our proxy adequately represents the sovereign wealth fund’s strategy. Taken together, these comparisons give us confidence that the simulated balance-sheet returns used in the main text are realistic and representative of actual public-sector portfolio behavior.



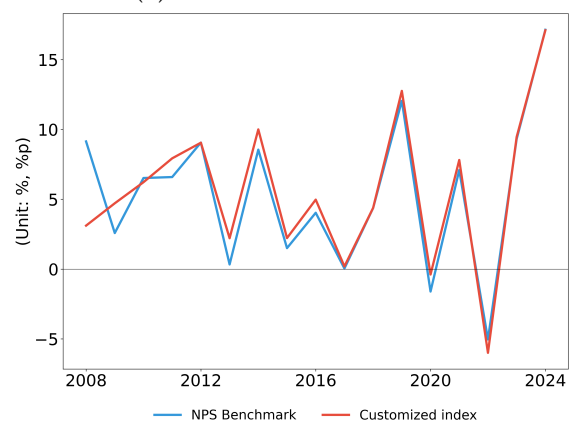
(a) Domestic Equity



(b) Domestic Fixed Income



(c) Global Equity



(d) Global Fixed Income

Figure 12: Comparison of NPS reported returns with proxy indices.

Source: NPS Annual Report; Bloomberg; Bank of Korea; Korea Investors Service (KIS).

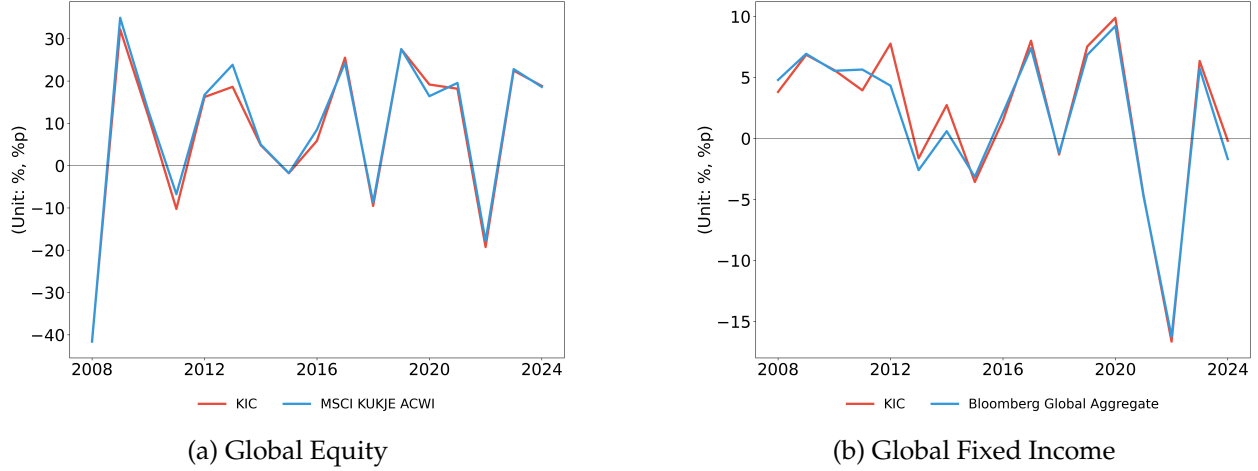


Figure 13: Comparison of KIC reported returns with proxy indices.

Source: KIC Annual Report; Bloomberg; Bank of Korea; Korea Investors Service (KIS).

E VAR Forecast

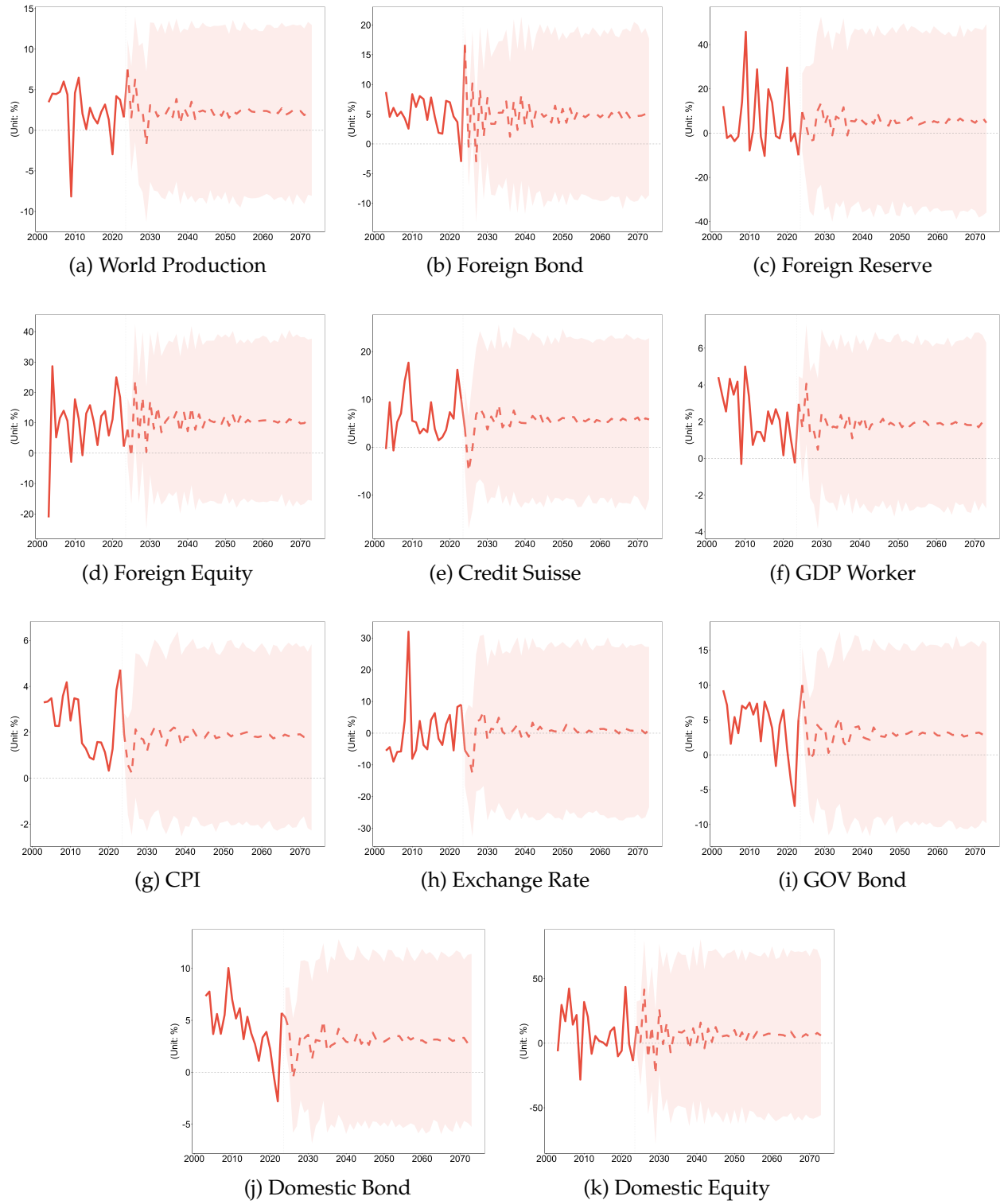
This appendix summarizes the stochastic forecasts for all 11 endogenous variables in our VAR system. The model is estimated with a lag length of 4, which was selected based on a standard information criterion.

Each panel in Figure 14 displays the median forecast path (solid red line) and the 95% confidence interval (the light-blue shaded fan) generated from 5,000 Monte Carlo simulations.

A key characteristic of the forecasts is their mean-reverting property. Because every series enters the VAR in a stationary form (as returns or first-differences), the forecast for each variable converges to its long-run unconditional mean over the projection horizon. Furthermore, the stabilization of both the median path and the width of the forecast band after an initial adjustment period confirms that the VAR model is stable, with all its characteristic roots (eigenvalues) lying strictly inside the unit circle.

To validate the predictive capability of our VAR model, we conducted an out-of-sample forecast accuracy test. The model was estimated using the first 80% of the sample and then used to generate forecasts for the remaining 20% hold-out period. The results indicate that the model performs reliably, with an average Root Mean Square Error (RMSE) of 9.62 and an average Mean Absolute Error (MAE) of 7.98 across all 11 endogenous variables. These metrics provide confidence in the model's stability and its suitability for the long-term forecasting exercise central to our analysis.

Figure 14: VAR Forecasting of Endogenous Variables (2024–2072)



Note: Shaded areas indicate the 95% confidence intervals; solid lines represent median values.

F Robustness: Bayesian VAR

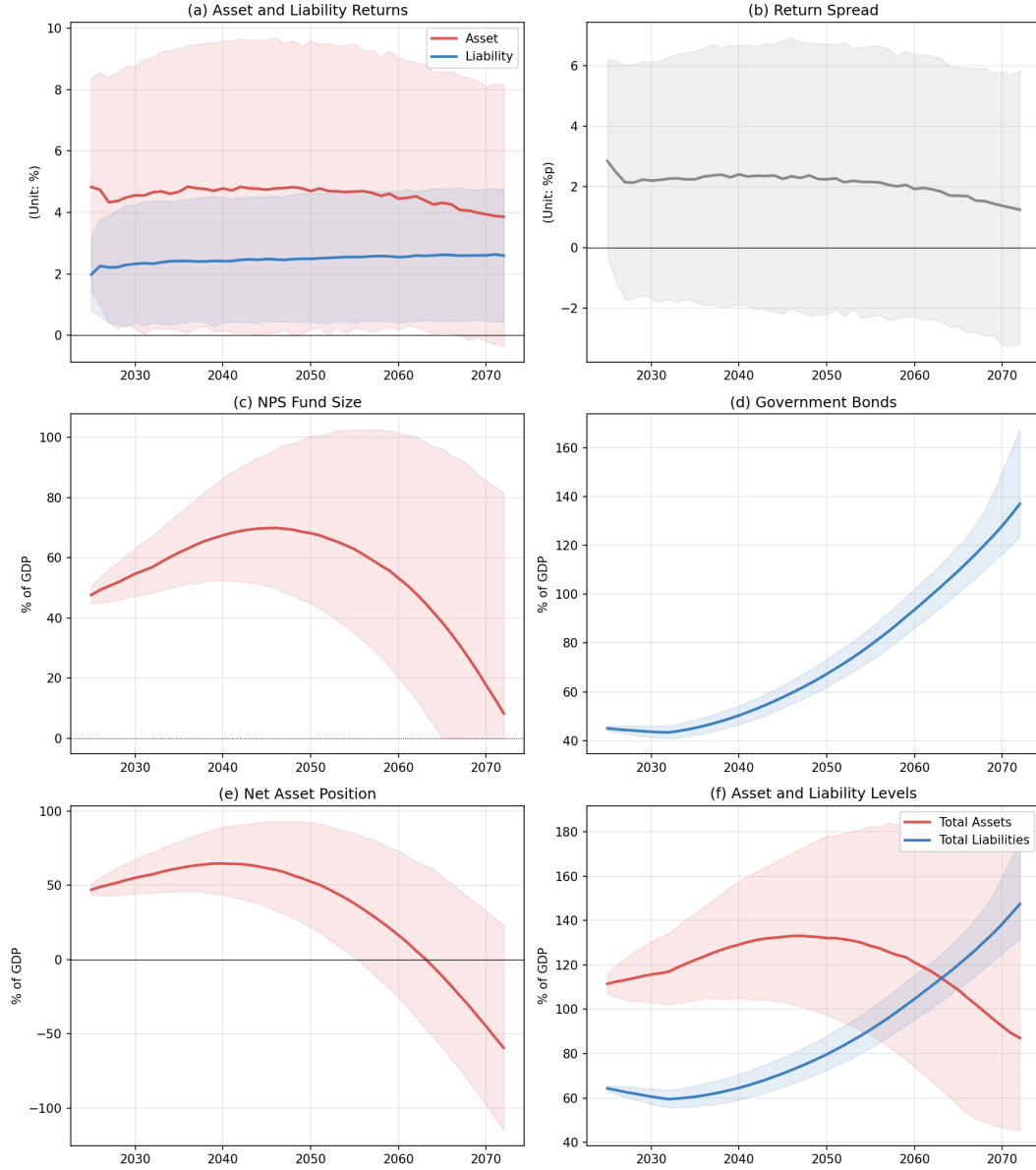
A practical concern with the unrestricted (OLS) estimates is that the largest eigenvalue of the estimated companion matrix is 0.94, very close to the unit circle. With only 88 quarterly observations, the long-run unconditional means of the system are therefore weakly identified, and the simulated paths inherit substantial estimation noise. The near-unit-root dynamics also generate a pronounced volatility drag: highly persistent, volatile return paths compound to a lower median level, which mechanically depresses the projected size of the NPS Fund and accelerates its depletion.

As a robustness check, we re-estimate the same eleven-variable system as a Bayesian VAR with a Minnesota prior. The prior shrinks each equation toward a white-noise specification (prior mean of zero on all autoregressive coefficients), with the standard hyperparameters governing overall tightness ($\lambda_1 = 0.2$), cross-variable shrinkage ($\lambda_2 = 0.5$), and the lag-decay rate ($\lambda_3 = 1$). This regularization pulls the dynamics away from the near-unit-root region: the largest eigenvalue of the posterior-mean system falls to 0.92, and the implied long-run mean of domestic equity returns rises from 5.9% (OLS) to 8.0% (BVAR).

Because the Bayesian estimates imply more stable and less volatile return paths, the volatility drag is smaller and the median NPS Fund is correspondingly larger. Under the BVAR, the projected return spread is modestly wider (approximately 2.1 percentage points, versus 1.9 under OLS) and the fiscal deterioration is slower: NPS depletion is pushed beyond the projection horizon and the terminal net asset position is less negative (approximately -59% of GDP, versus -76% under OLS). The two estimators thus bracket the result, but the qualitative conclusion is robust: under either specification the consolidated public sector runs a positive carry but is driven into a deeply negative net asset position by the structural deficits of an aging population. We adopt the more conservative OLS specification in the main text and report the BVAR as a regularized lower bound on the projected fiscal deterioration.

Figure 15 reproduces the main-text projections under the Bayesian VAR. The projected asset and liability returns (top panel) mean-revert to a slightly wider spread of about 2.1 percentage points, and the fiscal and pension indicators (middle panels) display the same dynamics as under OLS but with a slower deterioration: the median NPS fund peaks near 70% of GDP in the mid-2040s and remains positive through the horizon, while consolidated government bonds rise to about 137% of GDP by 2072 (against roughly 150% under OLS). The resulting net-asset and balance-sheet levels are shown in the bottom panels.

Figure 15: Projections under the Bayesian VAR (robustness)



Note: Bayesian VAR with a Minnesota prior ($\lambda_1 = 0.2$, $\lambda_2 = 0.5$, $\lambda_3 = 1$). Shaded areas indicate 95% confidence intervals; solid lines represent median values. Same setup as the main text otherwise.